Path-Gain Compressibility Determines Local Credit Assignment in Conductance-Based Dendrites

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Abstract

Biologically plausible credit assignment requires replacing the compartment-specific errors of backpropagation with signals available to dendritic synapses. We study when a low-dimensional somatic broadcast can serve as such a substitute in conductance-based dendritic trees. We derive exact gradients showing that each synaptic update factorizes into a local eligibility term, presynaptic activity times driving force times input resistance, and a single non-local compartment error. This exposes the key approximation made by local learning: replacing the exact path-specific compartment error with a broadcast field. We show that shunting inhibition changes this approximation by modulating input resistance along dendritic paths and, when it preferentially attenuates high-gain paths, compresses the effective path-gain field. Across matched additive and shunting dendritic networks, shunting improves local-backprop gradient alignment and reduces gradient scale mismatch. Transported-error oracle experiments show that supplying the exact path-gain-transported error largely closes the local-learning gap, identifying broadcast fidelity as the dominant bottleneck. Under nonnegative conductances and rank-1 broadcast, shunting LocalCA approaches matched backpropagation ceilings on MNIST, Fashion-MNIST, and context-gating, while routed and harder-data diagnostics reveal when richer pathway-specific feedback is required. These results identify path-gain compressibility as a mechanistic condition under which simple local credit assignment is effective in dendritic networks.

1 Introduction

Credit assignment in deep networks usually depends on backpropagation: global error transport through symmetric pathways with no obvious biological substrate. Dendritic compartments suggest an alternative. Each synapse has access to rich local state such as driving force, membrane conductance, and branch voltage, while supervisory information could arrive as a low-bandwidth broadcast from the soma [14]. The question here is not whether such a broadcast can match unconstrained backpropagation in all settings, but when it is a faithful approximation to the exact dendritic credit signal.

Starting from conductance-based dendritic voltage equations [1], we derive exact gradients for dendritic trees (Theorem 1) and show that each synaptic gradient factorizes into purely local eligibility terms and a single non-local compartment error. Approximating that compartment error with a broadcast signal yields a theorem-facing local rule (3F), practical optimizer wrappers (4F/5F), and higher-rank feedback variants for tasks where different branches require different teaching signals.

The central finding is mechanistic: shunting can make low-bandwidth broadcast useful by improving the compressibility of the conductance-stage path-gain field. Dendritic branching creates multiple error paths from distal synapses to the soma. Excitatory and inhibitory synapse banks set the conductance state of each

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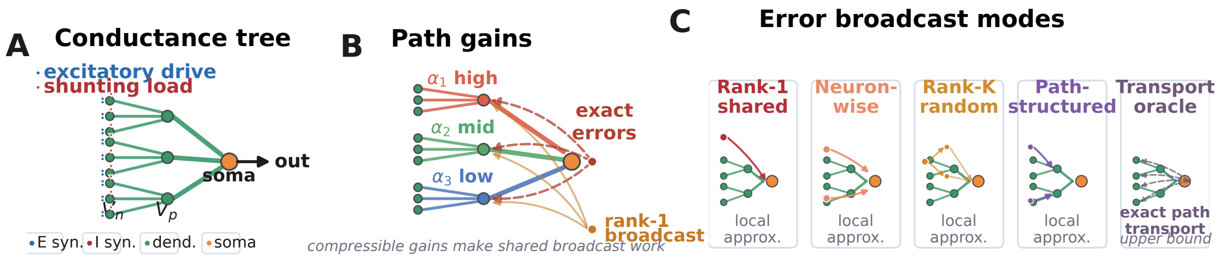

Figure 1: **Model and credit assignment.** **(A)** Conductance-based dendritic tree. Nonnegative excitatory and inhibitory inputs drive distal compartments, which feed proximal branches and then the soma; shunting inhibition enters the conductance denominator and lowers input resistance. **(B)** Path gains through the tree. Exact backprop supplies path-specific compartment errors, while rank-1 LocalCA replaces that field with one shared broadcast. **(C)** Error-broadcast choices. Rank-1, neuron-wise, rank-K, and path-structured broadcasts are local approximations with increasing bandwidth; transported error is an analysis oracle that supplies the exact path-specific signal.

compartment. Shunting makes inhibitory conductance enter the denominator, so inhibition changes input resistance and therefore the gain of every upstream path that passes through that compartment. This does not mean inhibition always helps. It helps rank-1 broadcast when it attenuates high-gain paths more than low-gain paths, narrowing or low-rank compressing the exact error field that the broadcast must approximate.

We quantify this chain with three diagnostics. First, exact-gradient reconstruction verifies that the mathematical factorization matches autograd to numerical precision. Second, gradient-fidelity measurements compare local and backprop gradients at matched parameters. Third, transported-error oracle experiments replace the broadcast with the exact path-gain-transported compartment error, isolating broadcast fidelity from other architectural factors. In a checkpointed MNIST gradient-dynamics diagnostic, shunting improves final local-backprop alignment (cosine 0.100 ± 0.033 vs. 0.005 ± 0.026) and reduces scale mismatch (0.26 ± 0.05 vs. 0.50 ± 0.05). On noise resilience, transported error raises additive local learning from 66.8% under rank-1 broadcast to 92.4–94.0%, nearly matching shunting at 94.3–95.2%.

Contributions. We make three claims: (i) exact conductance-tree credit factorizes into synapse-local eligibility and a path-specific compartment error (Theorem 1, Corollary 1); (ii) path-gain compressibility is a measurable mediator of when rank-1 broadcast approximates that compartment error, and shunting provides a biophysical route to such compression by changing input resistance along dendritic paths; and (iii) gradient-fidelity, transported-oracle, competence, and morphology experiments validate this mechanism while identifying the failure modes that require richer feedback.

The empirical story follows this sequence. Figure 1 defines the conductance tree, exact credit object, and broadcast taxonomy. Figure 2 verifies the exact factorization and measures local-backprop fidelity. Figure 3 links path-gain compressibility to broadcast fidelity, oracle transport, and learning. Figure 4 shows the practical supervised regimes. Morphology, harder-data, and routed-feedback experiments are reported as appendix diagnostics rather than headline benchmark claims.

2 Compartmental Voltage Model and Gradient Derivation

We use a steady-state conductance model derived from discretized passive cable dynamics [1, 2]. In normalized units, leak and inhibitory reversal are set to 0, the excitatory reversal is set to 1, and leak conductance is fixed at 1. Each compartment voltage is therefore a conductance-weighted average. Two

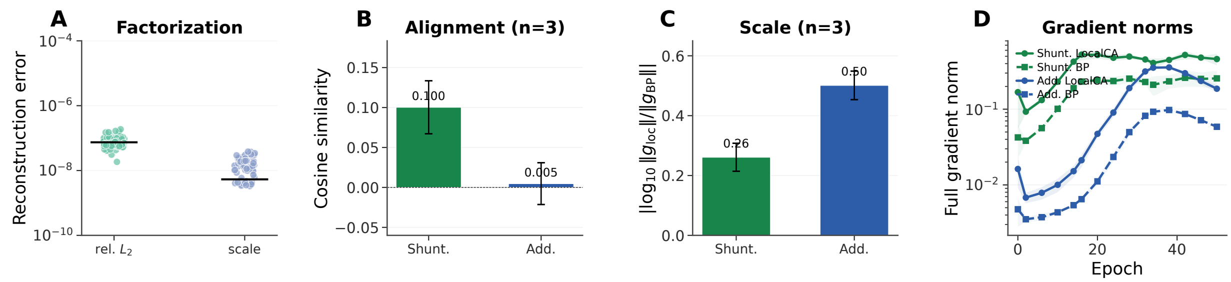

Figure 2: **Gradient-fidelity diagnostic.** **(A)** Exact-factorization sanity check across all theory-diagnostic conditions, reported as relative L_2 reconstruction error and scale mismatch. Reconstructed gradients match autograd to numerical precision. **(B)** Seeded final-checkpoint weighted cosine similarity between local and backprop gradients. **(C)** Seeded final-checkpoint scale mismatch, defined as $|\log_{10}(\|g^{\text{local}}\|/\|g^{\text{bp}}\|)|$. Error bars denote ± 1 s.d. across 3 checkpointed seeds. **(D)** Full-gradient norms over learning. Additive local and backprop gradients remain finite, so near-zero additive cosine reflects directional mismatch rather than vanishing gradients.

properties of this form matter for what follows: local sensitivities depend on both driving force $(E-V)$ and input resistance R^{tot}, and shunting inhibition corresponds to adding conductance when $E_{\text{inh}} \approx 0$. Throughout the paper we write dendritic morphologies as rooted trees $[b_1, b_2, \ldots, b_D]$, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. Thus $[3,3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma. In code, each excitatory unit is a stack of branch layers, each branch layer carries explicit excitatory and optional inhibitory synapse banks plus a branch-to-parent aggregation map, and the headline feedforward regime keeps `somatic_synapses=false` so inputs terminate on dendritic branches rather than directly on the soma. The main experiments use `input_mode=1`: the transfer layer supplies nonnegative excitatory and inhibitory input streams, and inhibition enters through learned branch-level I-to-E conductance banks. This is input-driven shunting inhibition, not a recurrent lateral inhibitory circuit. The additive and shunting cores are therefore architecture-matched at the tree level; they differ only in the branch-level voltage integration rule. These ingredients play distinct roles in the credit-assignment argument. Branching creates the path structure. E/I synapse banks determine the local conductance state. Shunting turns inhibitory conductance into a multiplicative change in input resistance. The local learning rule then asks whether a low-dimensional teaching signal can stand in for the exact path-specific error induced by that tree.

2.1 Voltage Equation and Local Sensitivities

Consider compartment n with synaptic inputs j (activity x_j, reversal E_j, conductance $g_j^{\text{syn}} \geq 0$) and dendritic inputs from children (voltage V_j, conductance $g_j^{\text{den}} \geq 0$). The steady-state voltage is:

$$V_n = \frac{\sum_j E_j x_j g_j^{\text{syn}} + \sum_j V_j g_j^{\text{den}}}{\underbrace{\sum_j x_j g_j^{\text{syn}} + \sum_j g_j^{\text{den}} + 1}_{g_n^{\text{tot}}}}, \qquad R_n^{\text{tot}} = 1/g_n^{\text{tot}}. \tag{1}$$

V_n is a convex combination of reversal potentials, child voltages, and leak. Writing $\mathscr{S}_n = \{0\} \cup \{E_j\}_j \cup \{V_j\}_j$, we have $\min \mathscr{S}_n \leq V_n \leq \max \mathscr{S}_n$ and $0 < R_n^{\text{tot}} \leq 1$. The local sensitivities follow directly:

Proposition 1 (Local Sensitivities)**.**

$$\frac{\partial V_n}{\partial g_i^{\text{syn}}} = x_i R_n^{\text{tot}}(E_i - V_n), \quad \frac{\partial V_n}{\partial V_i} = g_i^{\text{den}} R_n^{\text{tot}}, \quad \frac{\partial V_n}{\partial g_i^{\text{den}}} = R_n^{\text{tot}}(V_i - V_n). \tag{2}$$

2.2 Shunting Inhibition as Divisive Gain Control

An inhibitory synapse with $E_{\text{inh}} \approx 0$ contributes current $(0 - V_n) x_j g_j^{\text{syn}}$ and increases g_n^{tot}. Its sensitivity is $\partial V_n / \partial g_j^{\text{syn}} = -x_j R_n^{\text{tot}} V_n$, which corresponds to multiplicative attenuation (divisive normalization). Shunting is divisive at the voltage level, but its effect on firing rates can be subtractive in some regimes [7]. Inhibitory plasticity can balance excitation dynamically [8]; our learned inhibitory conductances serve an analogous role. The same denominator also gives the direct connection to path gain. Writing $G_n^E(x)$ and $G_n^I(x)$ for the total excitatory and inhibitory synaptic conductance impinging on compartment n,

$$R_n^{\text{tot}}(x) = \left(1 + G_n^E(x) + G_n^I(x) + G_n^{\text{den}}\right)^{-1}.$$

Increasing $G_n^I(x)$ lowers $R_n^{\text{tot}}(x)$ and therefore multiplicatively suppresses every upstream path whose error must pass through compartment n. In this sense inhibition can gate credit flow by changing the path gain, even when the inhibitory drive is feedforward rather than lateral. Prop. 2 formalizes this statement after the tree path gain is defined.

2.3 Exact Gradients for Dendritic Trees

Let V_0 be the somatic output and define the soma/core error as $\delta_0 := \partial L / \partial V_0$. For a linear decoder $\hat{y} = W_{\text{dec}} V_0$, this reduces to $\delta_0 = W_{\text{dec}}^\top (\partial L / \partial \hat{y})$; for a nonlinear decoder, δ_0 is the corresponding decoder-input Jacobian product.

Theorem 1 (Conductance-Stage Backpropagation on a Dendritic Tree)**.** *For a rooted dendritic tree with soma at node* 0 *and unique parent $p(n)$ for each non-somatic compartment, if each branch passes its pre-reactivation voltage directly to its parent, then the loss gradient satisfies*

$$\frac{\partial L}{\partial V_n} = \frac{\partial L}{\partial V_{p(n)}} R_{p(n)}^{\text{tot}} \, g_{n \to p(n)}^{\text{den}}, \tag{3}$$

with boundary condition $\partial L / \partial V_0 = \delta_0$. Because the graph is a tree, each compartment has a unique path to the soma. Defining the path gain

$$\alpha_n = \prod_{(i \to k) \in \text{path}(n \rightsquigarrow 0)} R_k^{\text{tot}} \, g_{i \to k}^{\text{den}}, \qquad \frac{\partial L}{\partial V_n} = \alpha_n \delta_0, \tag{4}$$

with $\alpha_0 = 1$.

Proof. Apply the chain rule on the tree-structured computation graph using Prop. 1. □

Proposition 2 (Inhibitory control of conductance-stage path gain)**.** *Let $\mathscr{A}(n)$ be the set of compartments on the path from compartment n to the soma, excluding n and including the parent compartments through which the error must pass. Holding dendritic coupling conductances fixed, an added inhibitory conductance $\Delta G_k^I(x) \geq 0$ at any $k \in \mathscr{A}(n)$ changes the conductance-stage path gain by*

$$\frac{\alpha_n(x; \Delta G^I)}{\alpha_n(x; 0)} = \prod_{k \in \mathscr{A}(n)} \frac{g_k^{\text{tot}}(x)}{g_k^{\text{tot}}(x) + \Delta G_k^I(x)}, \qquad \frac{\partial \log \alpha_n}{\partial G_k^I} = \begin{cases} -R_k^{\text{tot}}, & k \in \mathscr{A}(n), \\ 0, & k \notin \mathscr{A}(n). \end{cases} \tag{5}$$

Proof. Substitute $R_k^{\text{tot}} = 1/g_k^{\text{tot}}$ into the path-gain product in Eq. 4. Adding inhibitory conductance changes only the denominator at compartments on the path from n to the soma, which gives the product ratio and the log derivative. $\square$

Prop. 2 makes two directed predictions. First, inhibition is a path gate: it attenuates all credit paths below the inhibited compartment. Second, inhibition is useful for rank-1 local learning only when this attenuation makes the distribution of α_n more uniform or better aligned with the task. If inhibition is too strong, too heterogeneous, or generated by a poorly calibrated inhibitory population, it can suppress useful signals and hurt learning. In the additive control the same I input changes voltages and local eligibilities, but it does not enter the conductance-stage path recursion as a multiplicative resistance gate.

When does shunting compress path gains? The attenuation identity above is not by itself a concentration result. Let $z_n = \log \alpha_n$ be the log path gain for compartment n. If added inhibitory conductance contributes a path-dependent attenuation

$$\beta_n(x) = \sum_{k \in \mathscr{A}(n)} \log\left(1 + \frac{\Delta G_k^I(x)}{g_k^{\text{tot}}(x)}\right), \qquad z_n' = z_n - \beta_n, \tag{6}$$

then, over compartments or examples,

$$\mathrm{Var}(z') = \mathrm{Var}(z) + \mathrm{Var}(\beta) - 2\,\mathrm{Cov}(z, \beta). \tag{7}$$

Thus shunting narrows the log path-gain field exactly when

$$\mathrm{Cov}(z, \beta) > \frac{1}{2}\mathrm{Var}(\beta). \tag{8}$$

This identity follows by taking the logarithm of the product ratio in Prop. 2 and applying the variance identity for $z - \beta$. Eq. 8 is therefore a diagnostic, not an assumption: shunting narrows log path gains only when inhibitory attenuation is aligned with the high-gain paths. This is why the morphology and input-mode diagnostics are reported as operating-regime tests rather than as monotonic “more inhibition is better” claims.

Corollary 1 (Local–Global Factorization)**.** *The exact synaptic gradient at compartment n factorizes as:*

$$\frac{\partial L}{\partial g_i^{\text{syn}}} = \underbrace{x_i R_n^{\text{tot}} (E_i - V_n)}_{\textit{synapse-local eligibility}} \cdot \underbrace{\frac{\partial L}{\partial V_n}}_{\textit{compartment error}}, \tag{9}$$

where the eligibility term depends only on quantities available at synapse i (presynaptic activity x_i, input resistance R_n^{tot}, driving force $E_i - V_n$), and the compartment error $\partial L/\partial V_n$ is the sole non-local quantity.

Under identity upward transfer, this factorization isolates the only non-local quantity as $\partial L/\partial V_n = \alpha_n \delta_0$. With a post-voltage activation $a_n = f_n(V_n)$ transmitted to the parent, the exact recursion becomes $\partial L/\partial V_n = f_n'(V_n)\,(\partial L/\partial V_{p(n)})\,R_{p(n)}^{\text{tot}} g_{n \to p(n)}^{\text{den}}$, so the compartment error remains the sole non-local term but the effective path gain acquires local activation-derivative factors. Any practical local rule therefore depends entirely on how well a broadcast signal approximates that exact compartment error, which we evaluate in Sec. 4.2. In the models studied here, shunting improves this approximation by reducing and stabilizing R_n^{tot}, which narrows the spread of local sensitivities across the tree. The theorem is stated for the pre-reactivation compartment voltage V_n produced by the conductance stage itself. In the empirical networks, each dendritic branch layer may optionally apply a post-voltage reactivation $f(V_n)$ before passing activity to the next stage; unless otherwise noted, the manuscript sweeps use a learnable bounded `param_tanh` reactivation. Our exact-error diagnostics are computed on the hooked pre-reactivation voltage, so the factorization check and compartment-error analyses target the conductance-stage quantity defined above rather than the post-reactivation activity.

3 Local Learning Rules

3.1 Broadcast Error Approximation

We approximate the exact compartment error $\partial L/\partial V_n$ (Corollary 1) with a broadcast signal e_n derived from the somatic/core error δ_0. For a linear readout this reduces to $\delta_0 = W_{\text{dec}}^{\top}(\partial L/\partial \hat{y})$; for a nonlinear decoder we use the corresponding decoder-input Jacobian product $J_{\text{dec}}(h_{\text{core}})^{\top}(\partial L/\partial \hat{y})$ so that the broadcast remains defined in the same core-space coordinates. We distinguish five feedback objects:

- **Rank-1 shared broadcast:** $e_n = \bar{\delta} \cdot \mathbf{1}$, where $\bar{\delta} = \text{mean}(\delta_0)$ is one scalar field delivered to all compartments.
- **Neuron-wise broadcast:** $e_n = \delta_0$ when the layer width matches the core/output error dimension. In our headline feedforward settings this condition usually fails, so the default historical "per-soma" implementation reduces to rank-1 shared broadcast.
- **Rank-K random broadcast:** $c = P_K\delta_0 \in \mathbb{R}^K$ and $e_n = \sum_{k=1}^{K} q_{n,k}c_k$, where P_K is a fixed random projection and $q_{n,k}$ mixes those K channels into compartment n.
- **Path-structured broadcast:** $e_n = \alpha\,\bar{\delta}\cdot\mathbf{1} + (1-\alpha)\,\Gamma_n(\delta_0)$, where Γ_n projects the somatic error into router-inferred pathway roles, gates those roles by local pathway activity, and transports the resulting vector hierarchically through the block-structured dendritic tree.
- **Transported oracle:** $e_n^{\text{transport}} = \alpha_n\delta_0 = \partial L/\partial V_n$. This is an analysis upper bound, not a proposed local rule, because it supplies the exact path gain α_n from Theorem 1.

We use the historical `per_soma` implementation for most experiments, but in the main feedforward architectures it is effectively rank-1 because hidden widths exceed the decoder error dimension (Appendix Table S18). Under shunting, that deliberately low-bandwidth field is often sufficient on standard classification tasks. Routed tasks and rank-bridge controls in the appendix test the opposite regime, where rank-1 broadcast collapses branch identity and richer feedback is needed. A local-mismatch heuristic is included only as an appendix control because it performs much worse than the somatic-error broadcasts (Appendix A).

3.2 Three-Factor Rule (3F)

Definition 1 (3F Update)**.** *For synaptic and dendritic conductances:*

$$\Delta g_j^{\text{syn}} = \eta\,\langle x_j R_n^{\text{tot}}(E_j - V_n)\,e_n\rangle_B, \qquad \Delta g_j^{\text{den}} = \eta\,\langle R_n^{\text{tot}}(V_j - V_n)\,e_n\rangle_B, \tag{10}$$

where $\langle\cdot\rangle_B$ denotes the batch average.

The three factors are presynaptic activity x_j (or a voltage difference), postsynaptic modulation through driving force and input resistance, and the broadcast error e_n. The same rule applies to excitatory and inhibitory synapses. The sign difference comes from the driving force $(E_j - V_n)$.

Additive control. Rule (10) is the local gradient of the *shunting* voltage $V_n = \sum E_j g_j x_j / g_n^{\text{tot}}$. For the additive control ($V_n = \sum g_j x_j$, no divisive normalization), the correct local gradient is simpler: $\Delta g_j^{\text{syn}} = \eta\,\langle x_j\,e_n\rangle_B$, with no driving-force or R_n^{tot} terms ($R_n^{\text{tot}} \equiv 1$ by definition since there is no denominator). Throughout, each architecture uses the learning rule derived from its own forward-pass dynamics. This keeps the comparison architecture-matched rather than applying a shunting-derived rule to an additive model.

3.3 Practical Heuristic Wrappers: 4F and 5F

4F (heuristic attenuation proxy). In the conductance-stage recursion (4), the path gain α_n attenuates the somatic error differently at each compartment. To compensate without explicitly computing that gain, 4F uses an online correlation proxy

$$\rho_n = \mathrm{Cov}(\bar{V}_n, \bar{V}_0)/(\sqrt{\mathrm{Var}(\bar{V}_n)\,\mathrm{Var}(\bar{V}_0)} + \varepsilon).$$

High ρ_n up-weights branches whose voltages covary with the soma; low ρ_n down-weights branches where the broadcast is likely to be a poor proxy.

5F (confidence-weighted 4F). 5F adds a second heuristic factor

$$\phi_n = \mathrm{Var}(V_n)/(\sigma_{\mathrm{res}}^2 + \varepsilon),$$

where σ_{res}^2 is the residual variance of V_n after linear regression on the parent voltage. Operationally, ϕ_n is a branch-wise confidence gate: it becomes large when local voltages are strong and stable relative to unexplained residual fluctuations, and small when local fluctuations dominate. We clamp it to $[0.25, 4.0]$ for stability. This factor is empirical, not theorem-derived. The paper's core claim rests on the exact factorization, path-gain compressibility, and the transported-oracle experiment; the gain from ϕ_n should be read as practical denoising rather than theoretical necessity.

Proposition 3 (5F Update)**.**

$$\Delta g_j^{\mathrm{syn}} = \eta\, \rho_n \phi_n \langle x_j R_n^{\mathrm{tot}} (E_j - V_n)\, e_n \rangle_B. \tag{11}$$

A feedback-alignment-style random-broadcast argument is included in Appendix C; it is supportive but not central to the main mechanism claim.

3.4 Structured pathway feedback and branch-role plasticity

For tasks with latent pathway structure, scalar broadcast can erase the distinction between different dendritic branches that should be credited differently. Recent experimental evidence supports the existence of vectorized instructive signals in cortical dendrites [43], motivating the structured feedback variant we introduce here. We therefore augment the local rule with a structured cue-routing variant, *PV-LocalCA*.

Hierarchical pathway-vector broadcast. The encoder/router defines a low-dimensional role basis over latent pathways. Instead of collapsing δ_0 to a rank-1 or neuron-wise broadcast, we project it into this role basis and gate each role by local pathway activity:

$$e_n^{\mathrm{PV}} = \sum_{k=1}^{K} q_{n,k} c_k,$$

where $c_k = \langle r_k, \delta_0 \rangle$ are role coefficients and $q_{n,k}$ are local branch-role weights. Scalar broadcast is the special case $K{=}1$, while PV-LocalCA is a structured rank-K extension rather than a separate learning principle.

Router-derived branch roles. Rather than assigning hand-coded apical/basal labels, each branch j receives a normalized role profile r_j inferred from router assignments and incoming block weights. Plasticity is then gated by branch selectivity and by a synapse-specific role-alignment factor $\langle r_i, r_j \rangle$, biasing learning toward pathway-consistent afferents without imposing a fixed branch taxonomy.

4 Experiments

4.1 Setup

We evaluate two settings: a *capacity-calibrated* regime, where matched architectures achieve strong backprop performance, and *controlled sweeps* that isolate inhibition strength, broadcast mode, morphology, and architecture. The main text uses MNIST, Fashion-MNIST [37], context gating, and noise resilience; cue integration and CIFAR-10 are reported as appendix diagnostics because they are most useful for probing failure modes rather than establishing the core mechanism. Formal task definitions are in Appendix I. Architectures include point MLP baselines and dendritic cores with either additive integration or shunting. Main comparisons are architecture-matched additive vs. shunting cores and local vs. backprop training; DFA appears only as a supplementary baseline. Most MNIST, Fashion-MNIST, and context-gating results use 5F with the historical `per_soma` broadcast implementation. Because the hidden widths in these feedforward settings exceed the decoder output dimension, that implementation reduces to rank-1 shared broadcast in the main mechanism and competence sweeps; higher-rank controls then use explicit low-rank, path-propagation, path-transport, or pathway-vector variants. Unless otherwise stated, dendritic layers also use the learnable post-voltage reactivation `param_tanh`; Appendix Tables S11 and S12 then audit activation shape in a more controlled setting by comparing identity (`none`) against a frozen `param_tanh` under matched additive/shunting initialization. All dendritic runs supporting the main claims keep synaptic and dendritic conductances nonnegative via positive parameter transforms and enforce a nonnegative first-layer synaptic drive. Image inputs are used in $[0,1]$, noise-resilience corruption is clipped back to $[0,1]$, and context gating uses a nonnegative contextual figure-ground MNIST construction. The inhibition sweeps additionally use a transported-error oracle derived from Theorem 1 as an upper bound rather than as the proposed biological rule. Main mechanistic, oracle, competence, morphology, and activation-audit results use 5 seeds per condition unless otherwise noted. Appendix diagnostics state their seed counts in captions or tables.

The experiments follow three questions. First, does the exact factorization explain the gradient geometry observed in code? Second, does shunting change the path-gain and exact-error fields in regimes where rank-1 broadcast is expected to improve, and does exact path transport rescue learning when rank-1 broadcast is insufficient? Third, do the same ingredients remain useful in capacity-calibrated supervised tasks and across dendritic morphologies?

4.2 From Exact Factorization to Credit-Signal Fidelity

To test whether these performance differences correspond to better credit signals, we compare local and backprop gradients on the same batch at the same parameter values. For each parameter tensor p, we compute directional alignment (cosine similarity) and scale mismatch, defined as $|\log_{10}(\|g_p^{\text{local}}\|/\|g_p^{\text{bp}}\|)|$, aggregated by parameter count.

Across the checkpointed MNIST gradient-dynamics diagnostic, shunting is more aligned with backprop and closer in norm than the additive control at the final checkpoint: weighted cosine rises from 0.005 ± 0.026 to 0.100 ± 0.033, and scale mismatch falls from 0.50 ± 0.05 to 0.26 ± 0.05. Figure 2A confirms that the theory is operational in code under the nonnegative-input setup: reconstructing gradients from Corollary 1 using exact compartment errors yields weighted relative reconstruction error below 2×10^{-7} and weighted scale mismatch below 4×10^{-8} across runs. In a few low-norm parameter blocks, cosine becomes numerically unstable even though the reconstruction itself is essentially exact, so we report the norm-based diagnostics in the figure. Figure 2D rules out a trivial zero-gradient explanation: in the 5F trajectory diagnostic, additive local and backprop gradient norms remain finite throughout learning. The only approximation made by LocalCA is therefore the broadcast substitute for $\partial L/\partial V_n$.

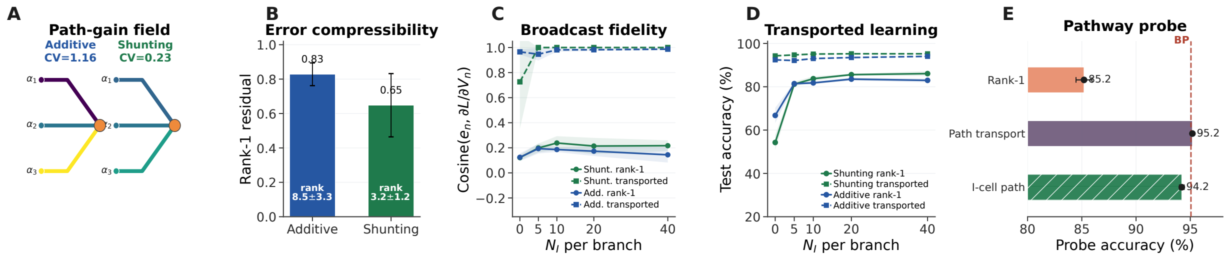

Figure 3: **Mechanistic chain from path gains to learning.** **(A)** Example dendritic path-gain field at matched inhibition, with measured MNIST CV annotated for additive and shunting networks. **(B)** Exact-error compressibility on MNIST: lower rank-1 residual and lower participation rank mean that a scalar broadcast is a better approximation. **(C)** Direct compartment-error fidelity on noise resilience. Solid lines compare rank-1 broadcast to the exact compartment error; dashed lines show the transported-error oracle. **(D)** Learning with the transported-error oracle. Improving the broadcast field recovers near-ceiling local learning, especially where rank-1 broadcast fails. **(E)** Compact inhibitory-path probe. In a one-layer noise-resilience test, exact path transport closes the rank-1 gap, and explicit inhibitory cells trained by the same local rule remain near the explicit-I backprop ceiling. Error bars denote ± 1 s.d. across seeds.

4.3 Mechanistic Chain: Path Gains, Broadcast Fidelity, and Oracle Transport

Figure 3 provides the paper's central empirical chain: shunting changes the dendritic path-gain field, makes the exact compartment-error field more compressible, improves broadcast fidelity, and then improves local learning when the transported-error bottleneck is removed.

Panel A visualizes the path-gain field on a representative dendritic tree. At matched inhibition (N_I=5), the measured MNIST path-gain CV is about 1.16 for additive integration and 0.23 for shunting, so the shunting tree has a much narrower conductance-stage gain range. Panel B measures compressibility directly on the exact compartment-error matrix. For MNIST, the best rank-1 residual falls from 0.83 ± 0.07 in additive networks to 0.65 ± 0.18 in shunting networks, and the participation-rank estimate falls from 8.5 ± 3.3 to 3.2 ± 1.2. This is the diagnostic matched to the paper's main claim: rank-1 broadcast works when the exact error field is closer to low rank.

Panel C then measures the sole non-local term from Corollary 1, $\partial L/\partial V_n$, directly. On the noise-resilience sweep, rank-1 broadcast tracks the exact compartment error better under shunting once inhibition is present: additive remains around 0.14–0.19 across the inhibited regimes, while shunting rises to about 0.20–0.24. The transported-error oracle obtained by recursively applying conductance-stage path gains improves fidelity sharply, reaching about 0.95–0.99 for additive and essentially 1.0 for shunting once inhibition is present.

Panel D trains the same local rule using that oracle signal. On noise resilience, transported broadcast lifts additive learning to about 92–94% and shunting to about 94–95% across inhibited regimes. On MNIST, transported broadcast reaches about 95.4–96.6% for additive and 96.5–97.3% for shunting. Appendix Fig. S9 reports the same ladder on flattened CIFAR-10 as a harder-data stress test: shunting rises from 29.7% under rank-1 broadcast to 47.2% under transported error, nearly matching its 49.5% backprop ceiling. The same sweep shows that learned inhibitory conductance is not a cosmetic addition: removing I-to-E conductance lowers shunting backprop from 49.5% to 43.9% and transported LocalCA from 47.2% to 38.6%. We use CIFAR-10 diagnostically rather than as a hard-vision benchmark claim.

Panel E keeps the inhibitory-path interpretation in the main figure. In a one-dendritic-layer noise-resilience probe, direct input-driven inhibition reaches only $85.2 \pm 0.7\%$ under rank-1 broadcast, but exact path transport reaches $95.2 \pm 0.0\%$. Replacing the direct inhibitory stream with explicit inhibitory dendritic cells trained by the same local rule remains near that ceiling ($94.2 \pm 0.2\%$). This does not make the explicit-I probe a headline benchmark, but it shows that an inhibitory population can implement the path-gain gating

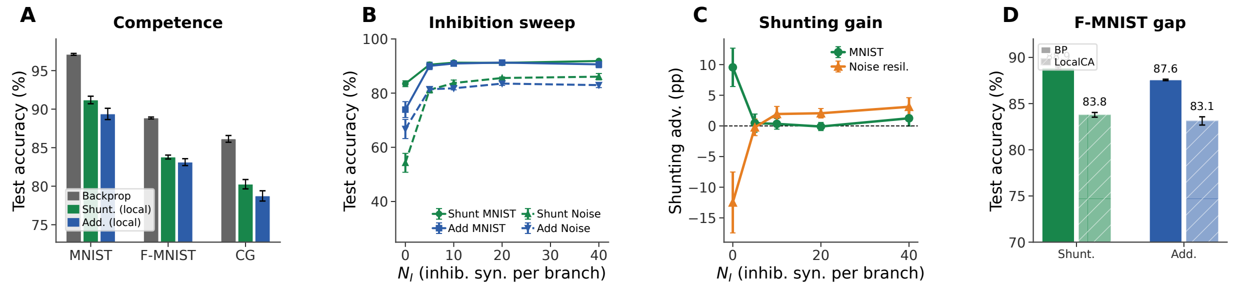

Figure 4: **Capacity-calibrated competence and regime dependence.** **(A)** Matched backprop ceilings vs. rank-1 5F LocalCA on MNIST, Fashion-MNIST, and context gating, including the matched additive context-gating rerun. **(B,C)** Inhibitory dose-response and shunting advantage on MNIST and noise resilience. **(D)** Fashion-MNIST confirms a smaller but consistent shunting advantage. Error bars denote ± 1 s.d. across seeds.

mechanism when the architecture isolates the dendritic-path question.

Appendix controls fill in the boundaries: coarse quantized broadcasts preserve performance until sign-only or sparse broadcasts remove too much information (Fig. S7); rank-K and path-propagation feedback close part of the transported-oracle gap (Table S10); additive normalization and DFA help but do not match the shunting conductance model (Figs. S11, S8); the full input-mode probe is reported in Fig. S13 and Table S8; and post-training inhibitory-conductance interventions plus exact-error SVD diagnostics show that learned inhibition is causally task-structured and makes the exact compartment-error field more compressible (Fig. S15).

4.4 Capacity-Calibrated Competence

Having established the mechanism, we next ask whether the same conditions improve end-task learning in architectures that are already competent under backpropagation. In the capacity-calibrated regime, shunting dendritic cores remain reasonably competitive under local learning: on MNIST, Fashion-MNIST, and context gating, the best local configuration (5F, rank-1 broadcast in these architectures, local decoder) stays within about 5–6 percentage points of the matched backprop ceiling (Table S4; Fig. 4). Within the local-rule family, 5F is the strongest practical wrapper, but it remains heuristic rather than theorem-derived; Appendix Tables S1, S3, and S2 summarize that division of labor across rules and broadcast choices. Classical random-feedback baselines remain much weaker on the same task family: in the matched MNIST appendix comparison, DFA reaches 66.1% on shunting and 62.2% on additive, well below the 91.4% best shunting LocalCA result, while FA is not directly compatible with the block-structured dendritic layers (Appendix Fig. S8). This helps separate the contribution of the conductance architecture from the contribution of the LocalCA rule itself.

4.5 Regime Dependence of the Shunting Advantage

The benefit of shunting varies across tasks. With matched rank-1 broadcast, shunting is modestly better on MNIST, Fashion-MNIST, and context gating; the matched additive context-gating run reaches 0.787 ± 0.007 versus 0.803 ± 0.006 for shunting. The gap grows on the inhibition-sensitive noise-resilience benchmark once inhibitory conductance is present (Fig. 4). This is the expected regime dependence: shunting helps most when input resistance is part of the useful credit geometry rather than merely an added conductance load.

Scope. The claim is mechanistic, not an unconstrained benchmark claim: CIFAR-10, routed feedback, morphology, and depth scaling are diagnostics of where rank-1 broadcast breaks, and shunting is not a universal solution for hard vision or biological learning.

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Dataset	Rule	Top-10 valid	Top-10 test
MNIST	3F	0.611	0.622
MNIST	4F	0.620	0.628
MNIST	5F	**0.912**	**0.916**
Context gating	3F	0.398	0.396
Context gating	4F	0.411	0.411
Context gating	5F	**0.807**	**0.789**

Table S1: **Rule-family ranking.** Top-10 mean across completed local-competence sweeps.

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A Supplementary Results

Mechanistic Support and Broadcast Controls

Rule-Family Ranking

Which Components Carry?

Component	**Representative effect**	**Interpretation**
Shunting architecture	In the nonnegative-input noise-resilience sweep, additive performs better at N_I=0 (0.668 vs. 0.543), but shunting overtakes it once inhibition is present (0.837 vs. 0.818 at N_I=10, 0.861 vs. 0.830 at N_I=40).	The shunting gain is operating-regime dependent rather than uniform: it appears once inhibition shapes conductance scaling strongly enough to help the shared broadcast.
Broadcast quality	On MNIST (5F), shunting per-soma reaches 0.912 ± 0.005, while local-mismatch remains at 0.146 ± 0.046.	Correctly matching the broadcast field to the somatic error matters far more than arbitrary local mismatch heuristics.
Transported oracle	Across inhibited noise-resilience regimes, additive local learning rises from roughly 81–84% under per-soma to 92–94% under transported broadcast; shunting rises from roughly 81–86% to 95–95.2%.	The oracle remains the strongest causal evidence that broadcast fidelity, not just architecture, is the dominant bottleneck.
Harder-data CIFAR-10 ladder	In the nonnegative-input CIFAR-10 control ladder, shunting backprop reaches $49.5 \pm 0.4\%$ vs. $48.3 \pm 1.0\%$ for additive; shunting transported LocalCA reaches $47.2 \pm 0.6\%$, while shunting rank-1 reaches $29.7 \pm 1.3\%$. Removing learned I-to-E conductance drops shunting transported LocalCA to $38.6 \pm 1.0\%$.	CIFAR-10 is not used as a benchmark claim, but it is a useful non-saturated stress test: the stronger result requires both shunting inhibition and a high-fidelity transported credit signal.
Rule hierarchy	In the competence sweeps, MNIST improves from 0.622 (3F) and 0.628 (4F) to 0.916 (5F); context gating improves from 0.396/0.411 to 0.789.	5F is the strongest practical rule in this paper, though its extra factor remains heuristic rather than theorem-derived.
5F stability	Tightening the ϕ clamp to $[0.5, 2.0]$ changes MNIST by only about $+0.4$ pp; varying EMA from 0.05 to 0.20 changes it by less than 0.2 pp.	The practical gain from 5F is not especially brittle to its main tuning knobs.
Structured feedback	In the cue-routing rank sweep, learned-routing additive local learning remains near ceiling while learned-routing shunting rises from 0.742 at random rank-1 to 0.846 at random rank-2; the current tuned pathway-vector variant reaches 0.815.	Routed tasks remain the main failure mode of rank-1 feedback, but the robust conclusion is that higher rank helps; the best structured fix is not yet settled enough to anchor a headline claim.
Random low-rank channels	On the shunting noise-resilience bridge, random low-rank broadcast improves from 0.453 at K=1 to 0.764 at K=4 and 0.795 at K=8, while exact path transport reaches 0.847; on cue routing, the best unstructured low-rank setting is K=2 at 0.846.	Higher-rank unstructured broadcast now closes a large fraction of the rank-1 gap, but it still does not recover the transported-oracle ceiling and does not by itself establish a structure-specific advantage.
Path propagation proxy	In the shunting noise-resilience bridge, the current morphology-aware path-propagation proxy reaches 0.762 ± 0.047, well above per-soma (0.662 ± 0.016) and close to random low-rank K=4 (0.764 ± 0.025), though still below exact path transport (0.847 ± 0.018).	The present recursive attenuation proxy is a meaningful constructive baseline: it closes much of the oracle gap, though a stronger strictly local approximation to transported error remains open.
Morphology regime	In the exploratory noise-resilience regime map (Fig. S12), depth-2 trees show their largest shunting gains near $N_I = 0$, whereas depth-3 trees retain their strongest gains around $N_I = 5$.	Morphology appears to shift the inhibitory operating regime in which shunting is most useful, suggesting that conductance normalization and tree geometry interact rather than contributing independently.
Activation choice	In the five-seed frozen activation audit, identity outperforms frozen `param_tanh` for LocalCA on noise resilience in both architectures (0.590/0.859 vs. 0.109/0.772 for additive at N_I=0/10; 0.411/0.769 vs. 0.331/0.701 for shunting), while the backprop effect remains strongly regime-dependent.	The post-voltage firing-rate transform is a real modeling variable, not just an implementation detail. Even a fixed saturating nonlinearity changes the operating point substantially, so the main paper's mechanistic claims should be read in the learned bounded `param_tanh` regime used throughout the headline experiments.

Table S2: **Which components carry the observed gains?** A compact summary of the empirical division of labor across architecture, broadcast design, and LocalCA rule variants.

Core	Broadcast	Decoder	Test (mean±std)
Shunting	per-soma	local	0.912 ± 0.005
Shunting	per-soma	backprop	0.909 ± 0.008
Shunting	local-mismatch	local	0.146 ± 0.046
Shunting	local-mismatch	backprop	0.146 ± 0.037
Additive	per-soma	local	0.894 ± 0.007
Additive	per-soma	backprop	0.900 ± 0.001
Additive	local-mismatch	local	0.342 ± 0.058
Additive	local-mismatch	backprop	0.348 ± 0.095

Table S3: **Local-mismatch recheck (MNIST, 5F).** Per-soma is consistently strong; local-mismatch remains much weaker.

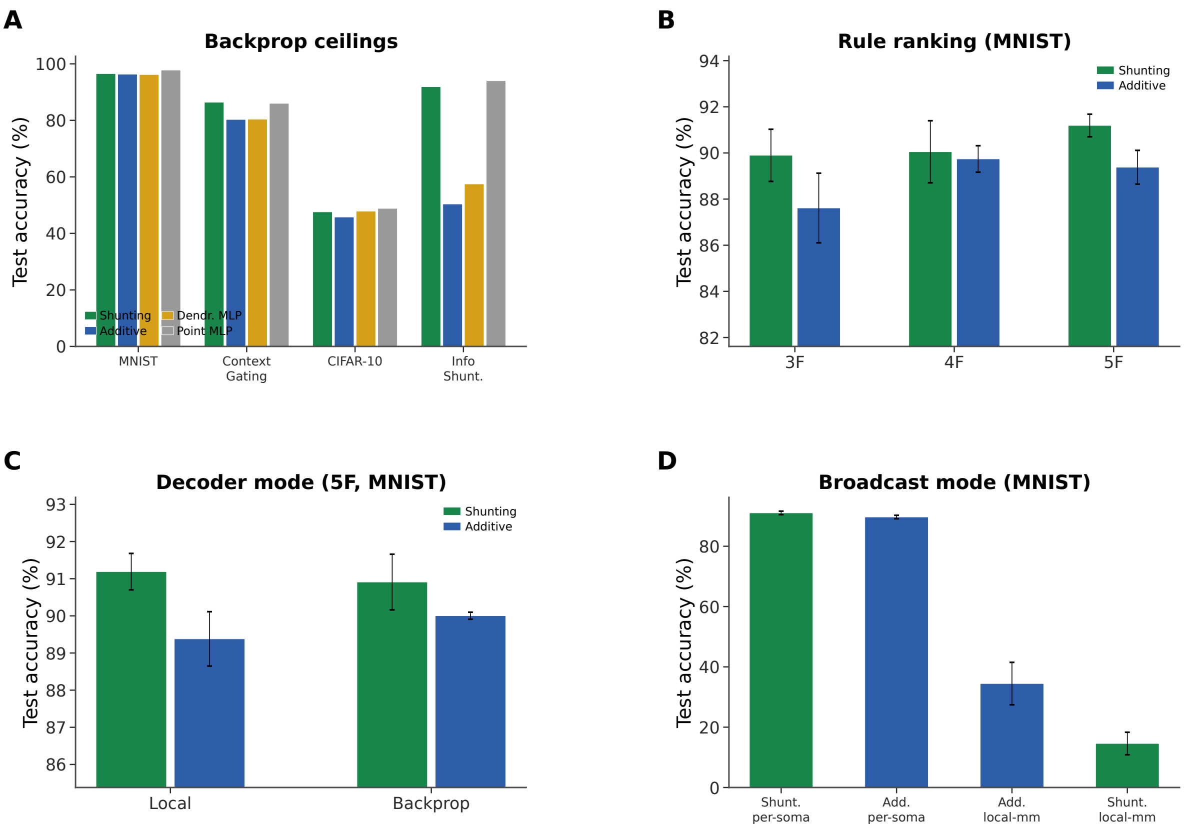

Figure S1: **Capacity calibration (supplementary).** **(A)** Phase 1 backprop ceilings across all architectures and datasets. **(B)** Rule-family ranking: 5F consistently outperforms 4F and 3F. **(C)** Decoder mode comparison (local vs. backprop decoder, 5F MNIST). **(D)** Broadcast mode comparison: per-soma is required for strong performance; local-mismatch fails.

Dataset	BP ceiling	Shunting local	Additive local	BP gap
MNIST	0.971	0.912 ± 0.005	0.894 ± 0.007	5.9%
Fashion-MNIST	0.889	0.838 ± 0.003	0.831 ± 0.004	5.1%
Context gating	0.861	0.803 ± 0.006	0.787 ± 0.007	5.8%

Table S4: **Local competence.** Backprop ceilings are matched-capacity shunting references from dedicated 5-seed standard-training refreshes for MNIST and context gating and from the dedicated 5-seed Fashion-MNIST competence sweep. Local values are 5F with rank-1 broadcast and local decoder updates. Context gating additionally uses an HSIC auxiliary objective (weight 0.01; Appendix E). Error bars on local values: ± 1 s.d. across 5 seeds.

Condition	Metric	Value
Decoder: local vs backprop vs frozen (MNIST, toy model)	test acc	0.379 vs 0.379 vs 0.176
Shunting vs additive, matched (MNIST, toy model)	Δ test	$+0.210$
Per-soma, path on vs off (synthetic task, toy model)	test / MI(E,I;C)	-0.003 / $+0.017$

Table S5: **Mechanistic ablations.** Results from controlled small-network experiments using minimal architectures to isolate individual factors. These runs use a small 2-layer dendritic network (branch factor [9]) trained for 30 epochs; they are not directly comparable to the headline capacity-calibrated results.

Broadcast Mode Comparison and Local-Mismatch Recheck

Competence, Verification, and Stress Tests

Phase 1 Capacity Ceilings

Local Competence and Regime Dependence

Ablation Results

Extended Gradient Analysis and N_I Sweep Detail

Alignment dynamics are not a zero-gradient artifact. To check whether low additive alignment could be caused by vanishing gradients, we analyzed a separate 3-seed checkpoint trajectory with parameter-level LocalCA and backprop gradients. The additive local gradients are nonzero throughout training (weighted norm range 9.5×10^{-4}–9.5×10^{-3}), and the corresponding backprop gradients are also nonzero (2.3×10^{-5}–5.7×10^{-3}). The low additive cosine therefore reflects directional and scale mismatch, not absent gradients: additive 5F cosine stays near zero from epoch 0 to epoch 50 (0.018 to 0.005), while its local/backprop norm ratio falls from about 8.8×10^{2} to 0.69.

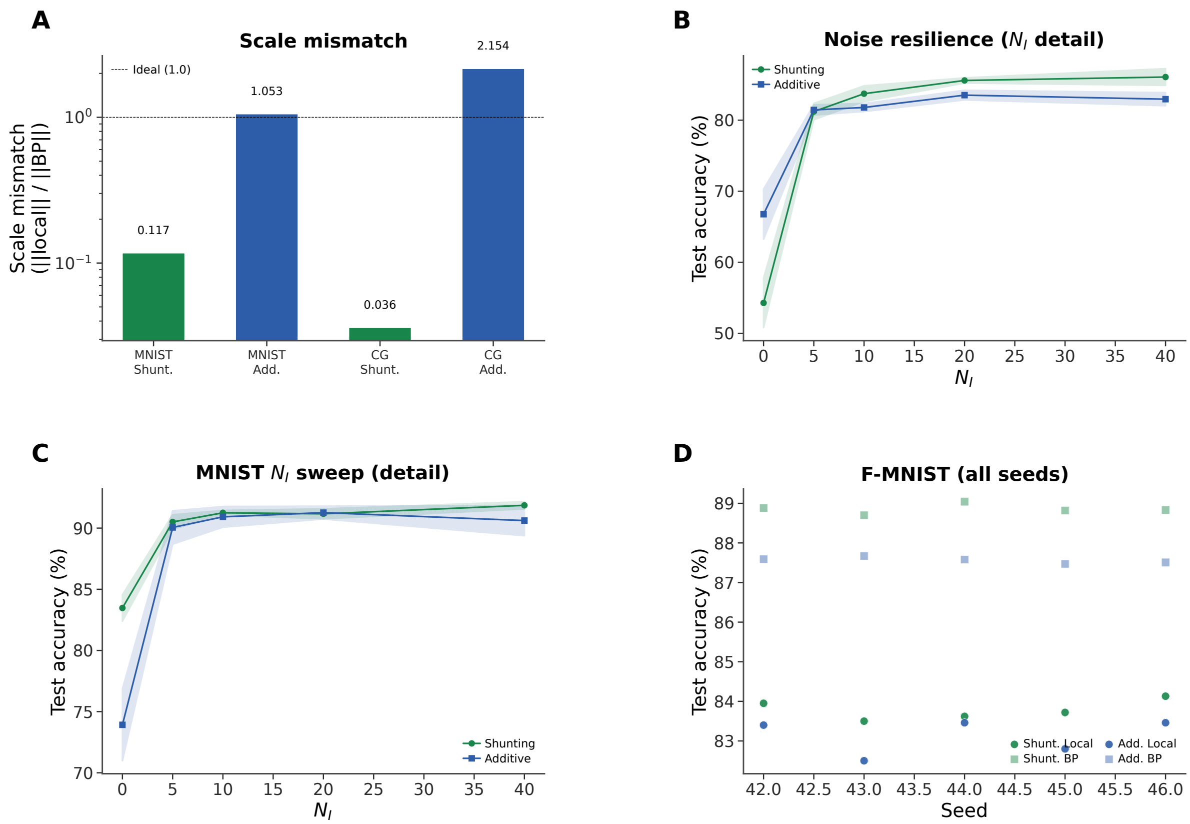

Figure S2: **Extended gradient and N_I analysis (supplementary).** **(A)** Scale mismatch bars: shunting achieves near-ideal scale (0.117); additive exhibits order-of-magnitude distortion (> 1.0). **(B)** Noise-resilience N_I dose-response with error bands (± 1 s.d.). **(C)** MNIST N_I dose-response detail with error bands. **(D)** Fashion-MNIST individual seeds for all conditions, showing consistency across runs.

Verification and Reproducibility

Additional Stress Tests

Broadcast Bandwidth (Quantization) Control

FA/DFA Baseline Comparison

CIFAR-10 Results

Table S6 provides the exact values for the appendix CIFAR-10 figure. In this stronger compact direct-I-stream family, shunting performs well under standard training (49.5% vs. 48.3% for additive), so the harder-data stress test does not unfairly penalize the shunting architecture. The broadcast-fidelity ladder also survives once decoder-aware soma mapping is used: additive LocalCA reaches 25.5% under rank-1 per-soma broadcast and 32.4% under transported error, while shunting rises from 29.7% under rank-1 broadcast to 35.0% with random low-rank K=4 and 47.2% with transported error, nearly matching the 49.5% shunting backprop ceiling. Removing learned I-to-E conductance lowers the shunting ceiling by 5.7 pp and transported LocalCA by 8.6 pp, making this the clearest harder-data evidence that inhibitory conductance is part of the useful credit geometry rather than just an added architectural detail.

A routed Double-CIFAR follow-up with strictly nonnegative router outputs was also not included as positive evidence. The task remained near chance even under matched backprop: additive BP reached

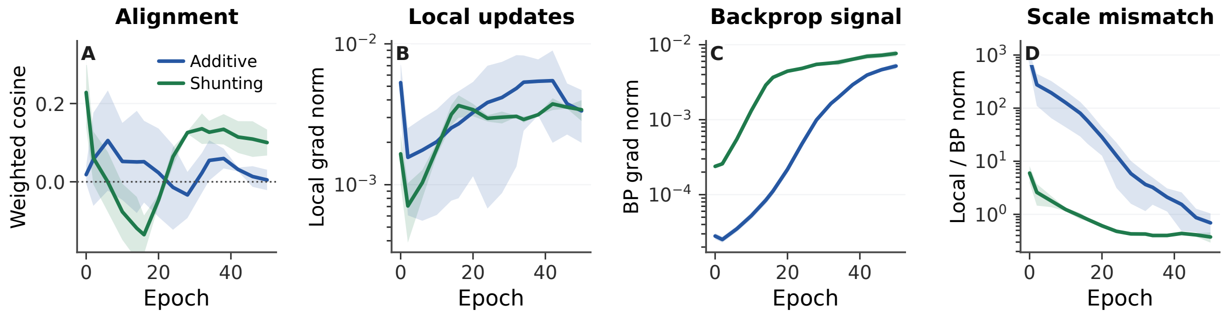

Figure S3: **Alignment dynamics with explicit gradient-norm checks. (A)** Weighted LocalCA-vs.-backprop cosine over training for the 5F rule. **(B,C)** Weighted local and backprop gradient norms stay nonzero for both additive and shunting models. **(D)** The additive control begins with a much larger scale mismatch, then approaches the backprop scale without recovering strong directional alignment. Shading is s.d. across 3 seeds.

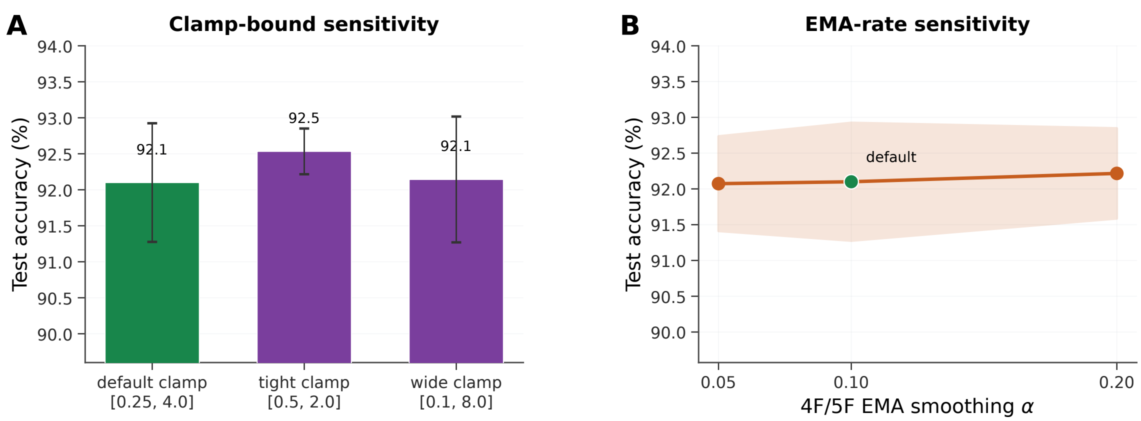

Figure S4: **5F sensitivity analysis (MNIST, shunting, per-soma broadcast). (A)** Sensitivity to the ϕ_n clamp interval. The default clamp $[0.25, 4.0]$ is already strong $(92.1 \pm 0.8\%)$; tightening it to $[0.5, 2.0]$ slightly improves performance $(92.5 \pm 0.3\%)$, while widening it to $[0.1, 8.0]$ is effectively neutral $(92.1 \pm 0.9\%)$. **(B)** Sensitivity to the 4F/5F EMA rate used for online statistics. Varying α from 0.05 to 0.20 changes mean test accuracy by less than 0.2 pp. Together these results suggest that the 5F gain is not driven by a brittle single hyperparameter choice, even though the factor itself remains heuristic. Errors: ± 1 s.d. across 3 seeds.

$10.9 \pm 0.7\%$, shunting BP with learned I-to-E reached $11.6 \pm 0.4\%$, and the strongest LocalCA condition, shunting transported error with learned I-to-E, reached only $11.7 \pm 0.3\%$ (Fig. S14D). We therefore treat this as a failed hard routed-task screen and keep the stronger compact CIFAR-10 control ladder as the paper's harder-data diagnostic.

CIFAR-10 soma-on extension. We also tested the soma-on extension on the same compact direct-I-stream depth-4 family with quantile init (Fig. S10; Table S7; 3 seeds). With `somatic_synapses=true`, path-transported LocalCA stays strong in both cores: additive reaches $45.5 \pm 0.6\%$ and shunting reaches $46.6 \pm 1.0\%$. Relative to the soma-off control ladder, the additive path-transport condition improves substantially, while shunting remains within 1 pp of its soma-off path-transport value (47.2%) and within 3 pp of the matched shunting backprop ceiling (49.5%). Per-soma rank-1 broadcast also becomes much more competitive for additive $(41.6 \pm 0.6\%)$ than for shunting $(35.9 \pm 1.2\%)$, while low-rank $K{=}4$ remains weak in both cores.

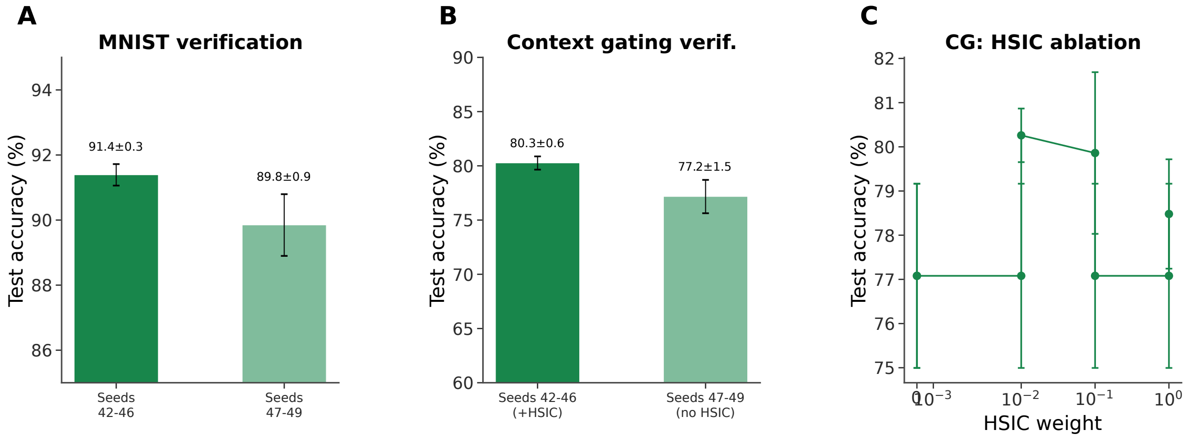

Figure S5: **Verification and reproducibility (supplementary).** **(A)** MNIST verification: main seeds (42–46) yield $91.4 \pm 0.3\%$; held-out seeds (47–49) yield $89.8 \pm 0.9\%$, confirming generalization. **(B)** Context gating verification: main seeds (+HSIC) $80.3 \pm 0.6\%$; held-out seeds (no HSIC) $77.2 \pm 1.5\%$. The roughly 3 pp gap reflects HSIC removal, not seed sensitivity. **(C)** HSIC weight ablation on context gating: moderate weights (0.01–0.1) perform best.

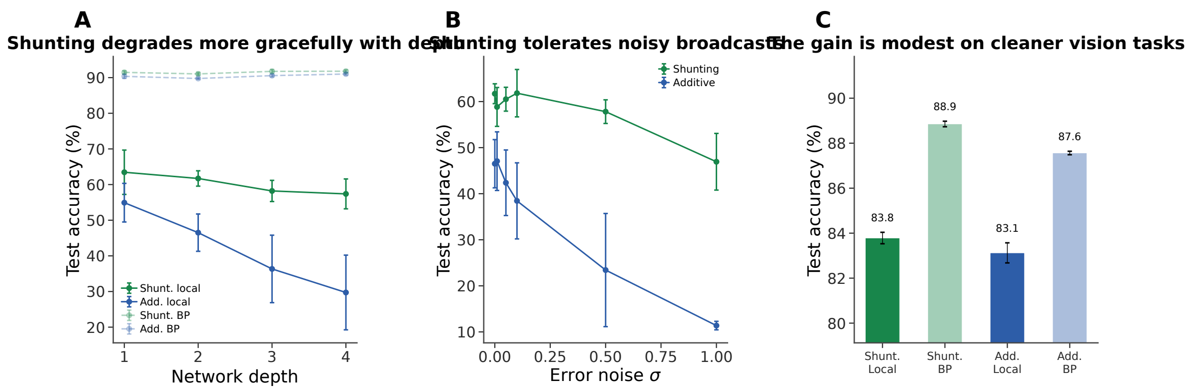

Figure S6: **Additional stress tests.** **(A)** Depth scaling: shunting local (green) degrades more gracefully from 63.5% to 57.4%; additive local (blue) drops from 54.9% to 29.7%. Backprop ceilings (dashed) remain stable. **(B)** Broadcast-noise robustness: shunting remains stable for $\sigma \leq 0.1$; additive degrades rapidly and reaches chance at $\sigma = 1$. **(C)** Fashion-MNIST: the shunting gain is modest on this cleaner benchmark (83.8% local vs. 88.9% backprop), consistent with the regime-dependent story in the main text.

Figure S10 makes the two useful views explicit: the absolute soma-on ordering still favors path transport strongly, and shunting transported error is already near its soma-off ceiling. On this harder dataset, the soma channel helps, but it still does not eliminate the importance of broadcast quality.

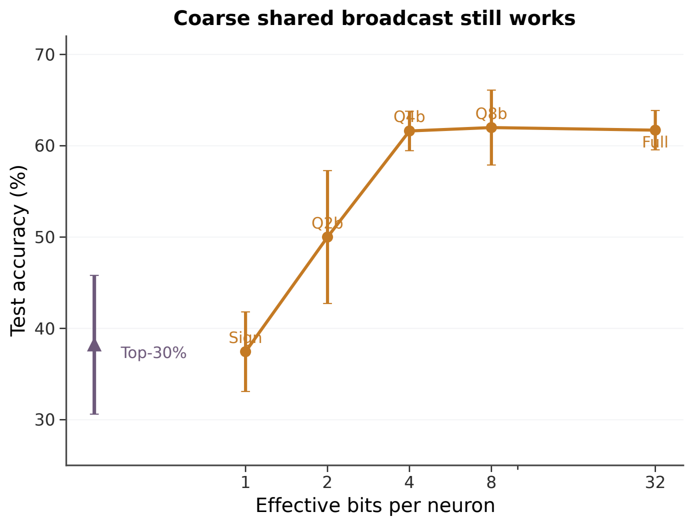

Figure S7: **Coarse shared broadcast still works.** Test accuracy as a function of effective broadcast bandwidth. Full-precision, 8-bit, 4-bit, and 2-bit quantization of the shared error signal all reach essentially the same accuracy (61–62%; $p > 0.9$ for 4-bit vs. full), while a sign-only broadcast and a sparse top-30% broadcast degrade sharply. This shows that the shunting LocalCA signal carries most of its useful information in its direction rather than in fine-grained magnitude, consistent with the mechanism-summary picture in the main text. This panel is reported separately so the main figure can carry the full causal chain (path-gain compressibility $\rightarrow$ alignment $\rightarrow$ oracle $\rightarrow$ global summary).

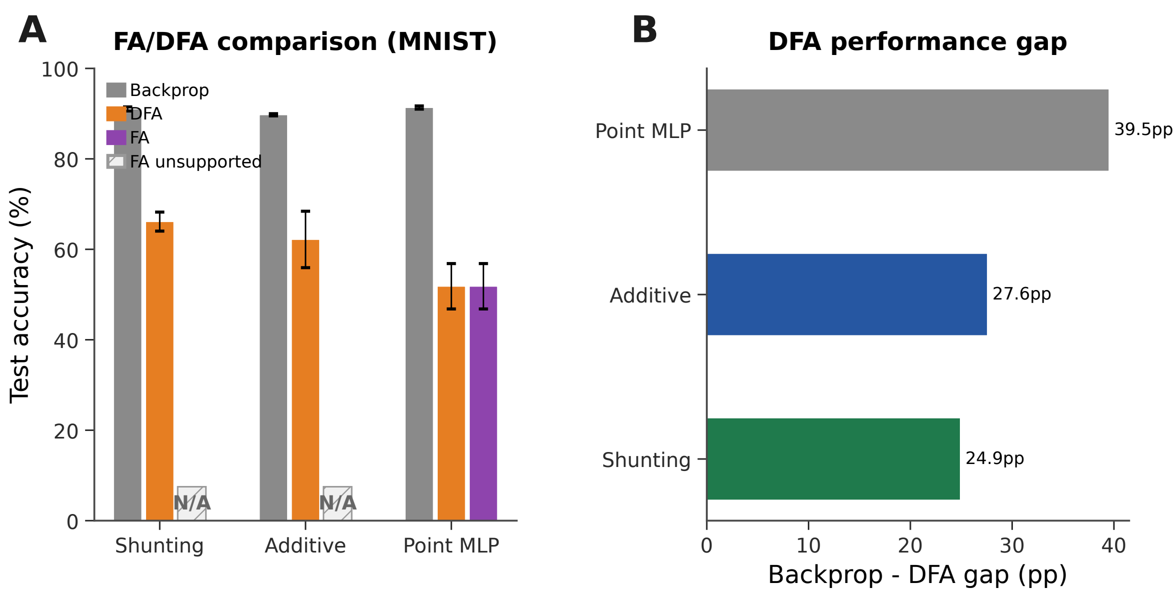

Figure S8: **Feedback alignment baselines (MNIST). (A)** Grouped comparison: standard backprop (neutral gray), DFA (orange), and FA (purple). Hatched N/A markers indicate that FA is not implemented for the block-structured dendritic architectures because the random feedback matrices are not dimensionally compatible with the tree-structured branch updates. DFA achieves 66.1% on shunting vs. 62.2% additive vs. 51.8% point MLP. **(B)** Backprop$-$DFA gap: dendritic architectures (24.9–27.6 pp) show smaller gaps than point MLPs (39.5 pp), suggesting conductance-based architecture is partially compatible with random feedback.

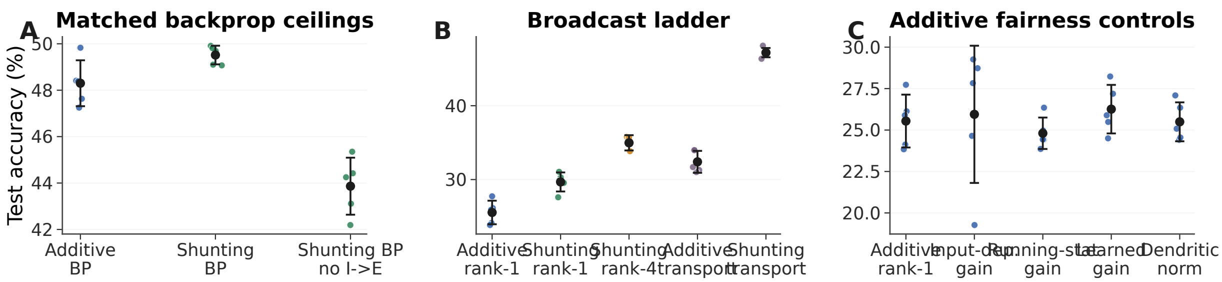

Figure S9: **CIFAR-10 control ladder in the compact direct-I-stream family.** **(A)** Matched backprop ceilings. With nonnegative inputs and a nonlinear decoder, shunting slightly exceeds additive under standard training, and removing learned I-to-E conductance lowers the shunting ceiling. **(B)** Broadcast ladder. Rank-1 broadcast remains weak on this harder dataset, rank-4 improves partially, and exact transported error nearly reaches the shunting backprop ceiling. **(C)** Additive fairness controls. Input-dependent gain, running-stat gain, learned gain, and dendritic normalization do not rescue additive rank-1 broadcast. This figure is a harder-data diagnostic, not a claim of competitive CIFAR-10 benchmarking.

Condition	Core	Training / broadcast	I-to-E	Test acc.
standard	additive	backprop	yes	48.3 ± 1.0
standard	shunting	backprop	yes	$\mathbf{49.5 \pm 0.4}$
standard, no learned inhibition	shunting	backprop	no	43.9 ± 1.2
rank-1	additive	5F per-soma	yes	25.5 ± 1.6
rank-1, input-dependent gain	additive	5F per-soma	yes	25.9 ± 4.1
rank-1, running-stat gain	additive	5F per-soma	yes	24.8 ± 0.9
rank-1, learned gain	additive	5F per-soma	yes	26.3 ± 1.5
rank-1, dendritic normalization	additive	5F per-soma	yes	25.5 ± 1.2
rank-1	shunting	5F per-soma	yes	29.7 ± 1.3
rank-1, no learned inhibition	shunting	5F per-soma	no	22.3 ± 0.7
transported error	additive	5F path transport	yes	32.4 ± 1.5
rank-4	shunting	5F low-rank	yes	35.0 ± 1.0
transported error	shunting	5F path transport	yes	$\mathbf{47.2 \pm 0.6}$
transported error, no learned inhibition	shunting	5F path transport	no	38.6 ± 1.0

Table S6: **Exact CIFAR-10 values for the 70-run control ladder (5 seeds per condition, validation-best epoch).** All rows use flattened CIFAR-10 in $[0,1]$, nonnegative transfer outputs, positive conductance transforms, a compact depth-4 dendritic core, and a nonlinear decoder. LocalCA maps output error back to decoder-input/soma coordinates through the decoder Jacobian. The useful conclusions are narrow but important: learned shunting slightly improves the matched backprop ceiling, learned I-to-E conductance is necessary for the strongest shunting results, transported error nearly closes the shunting LocalCA gap, and additive rank-1 fairness controls do not explain the transported-shunting result.

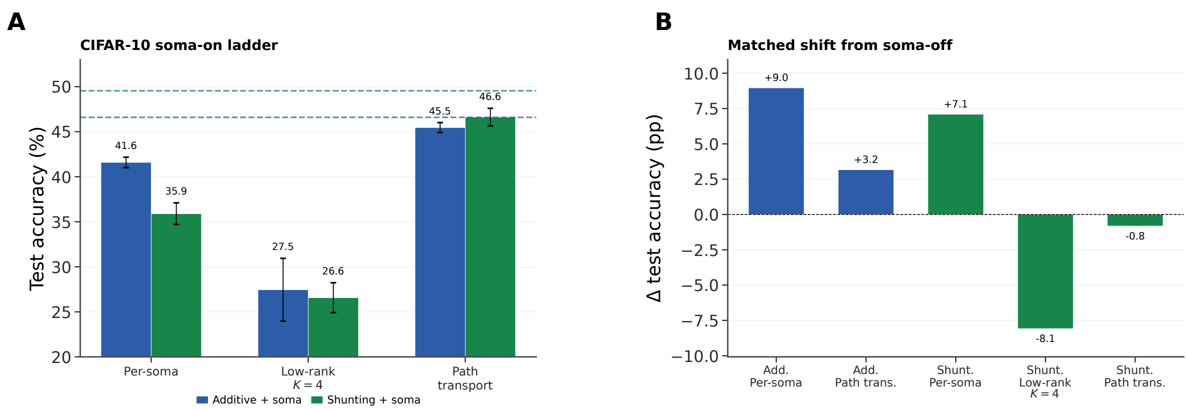

Figure S10: **CIFAR-10 compact direct-I-stream depth-4 with `somatic_synapses=true`. (A)** Absolute soma-on ladder on the compact direct-I-stream depth-4 family. Path transport stays strongest in both cores and remains near the matched backprop ceilings (dashed lines), while low-rank K=4 remains weak. **(B)** Matched shift from soma-off to soma-on for the conditions with a clean paired baseline. Soma helps additive strongly under per-soma and path transport, improves shunting per-soma, leaves shunting transported error close to its soma-off value, and worsens shunting low-rank K=4. Additive low-rank K=4 is omitted from panel B because the soma-off mechanism package did not include that exact matched condition.

Condition (soma-on)	**Additive**	**Shunting**
LocalCA per-soma (rank-1)	41.6 ± 0.6	35.9 ± 1.2
LocalCA low-rank K=4	27.5 ± 3.5	26.6 ± 1.7
LocalCA path transport	45.5 ± 0.6	$\mathbf{46.6 \pm 1.0}$
soma-off path transport	32.4 ± 1.5	47.2 ± 0.6
standard BP (ceiling)	48.3 ± 1.0	49.5 ± 0.4

Table S7: **CIFAR-10 compact direct-I-stream depth-4 with `somatic_synapses=true`.** Adding direct somatic synapses and using quantile reactivation init substantially improves the additive transported-error condition and keeps the shunting transported-error condition within 1 pp of its soma-off counterpart. Per-soma (rank-1) becomes much more competitive for additive than for shunting, while low-rank K=4 remains weak in both cores. The soma channel therefore helps, but does not remove the need for a strong credit signal.

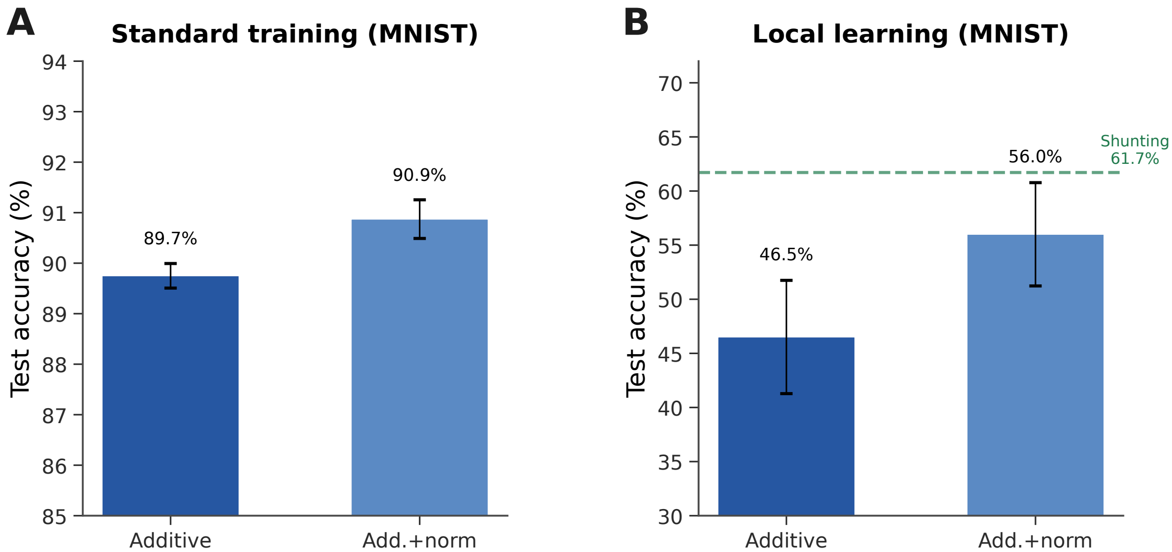

Figure S11: **Additive + normalization control (MNIST).** **(A)** Standard backprop: normalization provides a small boost (89.7% $\rightarrow$ 90.9%). **(B)** Local learning: normalization improves additive from 46.5% to 56.0% (+9.5 pp), partially closing the gap to shunting (61.7%, dashed green). The remaining gap (-5.7 pp) indicates that shunting provides benefits beyond simple voltage normalization.

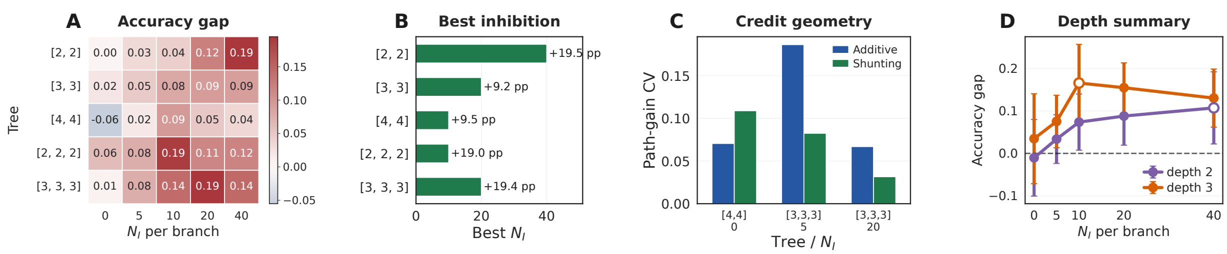

Figure S12: **Morphology shifts the useful inhibitory regime.** **(A)** Shunting-minus-additive LocalCA accuracy gap across dendritic morphologies and inhibitory synapse counts. **(B)** Best inhibitory count for each morphology. **(C)** Representative path-gain diagnostic for selected morphology cells, used to connect the performance map to credit geometry. **(D)** Average shunting advantage by depth, with markers showing the peak inhibitory count for each depth.

Additive Normalization Control

Structured Feedback, Activation, and Supportive Extensions

Morphology $\times$ Inhibition Regime Map

This supportive regime map tests whether tree geometry changes the inhibitory operating point at which shunting helps. We treat it as evidence that morphology changes the useful inhibition regime; a fuller mechanistic account would still require matched theory diagnostics for each cell in the grid.

Input-Mode Probe: Direct Inhibitory Stream vs. Explicit Inhibitory Cells

The headline experiments use `input_mode=1`, where the transfer layer provides nonnegative excitatory and inhibitory streams from the external input and the excitatory dendrites receive learned I-to-E conductance

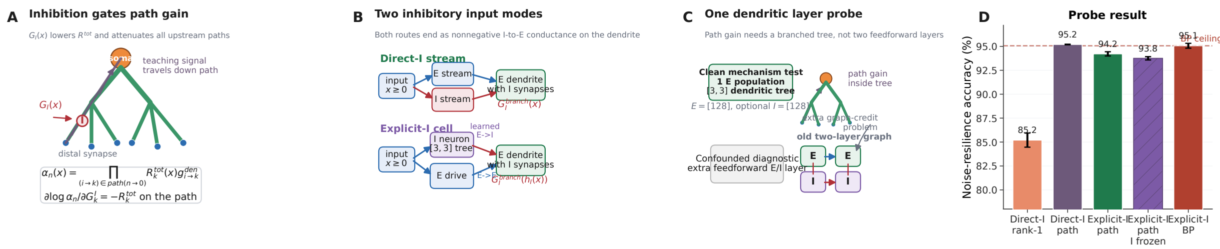

Figure S13: **Pathwise inhibition story and one-layer input-mode probe. (A)** In a branched conductance tree, the error path gain α_n is the product of local input resistances and dendritic coupling conductances along the path to the soma. Inhibitory conductance on any compartment along that path lowers local input resistance and attenuates all upstream credit paths. **(B)** The same branch-level inhibitory conductance can be supplied directly from a nonnegative inhibitory input stream (`input_mode=1`) or through an explicit feedforward inhibitory cell (`input_mode=0`). **(C)** The clean mechanism probe keeps the dendritic tree depth ($[3,3]$) but uses only one feedforward dendritic population; adding a second feedforward E/I layer tests a deeper graph-credit problem rather than the path-gain mechanism itself. **(D)** Noise-resilience results for the one-layer probe. Direct-I path transport reaches the explicit-I backprop ceiling, and explicit inhibitory cells with path-transport LocalCA also remain near ceiling. This supports the interpretation that explicit inhibitory cells can generate useful path-gain structure when the architecture isolates the dendritic-path question.

banks directly. To test whether the same mechanism can be generated by an explicit feedforward inhibitory population, we ran a one-feedforward-layer noise-resilience probe that preserves the relevant dendritic depth ($[3,3]$ morphology) while removing an unnecessary second feedforward population. In this clean setting, explicit inhibitory cells reach $94.2 \pm 0.2\%$ under path-transport LocalCA when their dendrites use the same update policy as excitatory dendrites (`explicit_inhibitory_update_mode=local_ca`), $93.8 \pm 0.2\%$ when those inhibitory-cell updates are frozen (`freeze`), and $95.1 \pm 0.3\%$ under standard backprop. Direct input-driven inhibition gives the expected broadcast-fidelity ladder on the same one-layer architecture: per-soma rank-1 reaches $85.2 \pm 0.7\%$, while exact path transport reaches $95.2 \pm 0.0\%$. Thus explicit inhibitory cells can generate useful path-gain structure once the architecture isolates the dendritic-path question. The older two-feedforward-layer explicit-I diagnostic was much weaker (about 58–63% under LocalCA despite a 94.7% backprop ceiling), so we treat it as evidence that deeper explicit $E \to I \to E$ graphs need graph-aware local errors, not as evidence against inhibition-mediated path gains.

Fitted inhibitory path-gain diagnostic. We also asked whether the direct-I-stream models actually use inhibitory synapses in the way implied by Prop. 2. We reran the cue-routing pathway controls with strictly nonnegative router outputs and matched learned-I versus no-I conditions (50 complete runs). The task is now too easy to separate feedback methods: all conditions reach 99.0–99.1% test accuracy, and removing learned I-to-E conductance changes accuracy by less than the seed-to-seed variation. The useful information is therefore mechanistic rather than performance-based. In the learned-I LocalCA runs, the first-layer pathway-inhibition balance carries context information, and higher inhibitory conductance is consistently associated with lower local input resistance. The sign of the balance can flip across random seeds, so we report best-polarity context decoding as a sign-invariant measure. This does not prove that pathway-vector feedback has solved routed credit assignment, but it supports the narrower claim that learned inhibitory conductance can create input-dependent path-gain differences by modulating branch resistance.

Causal inhibition and exact-error rank. We next tested whether learned inhibitory conductance carries task-specific structure rather than only changing global scale. We replayed trained shunting LocalCA

Inhibitory input mode	Training / broadcast	Test accuracy
Direct inhibitory stream (`input_mode=1`)	LocalCA, per-soma rank-1	0.852 ± 0.007
Direct inhibitory stream (`input_mode=1`)	LocalCA, exact path transport	$\mathbf{0.952 \pm 0.000}$
Explicit I cells (`input_mode=0`)	LocalCA, path transport, I-cell updates on	0.942 ± 0.002
Explicit I cells (`input_mode=0`)	LocalCA, path transport, I-cell updates frozen	0.938 ± 0.002
Explicit I cells (`input_mode=0`)	standard backprop	0.951 ± 0.003

Table S8: **One-feedforward-layer input-mode path-gain probe on noise resilience (3 seeds).** All rows use one excitatory dendritic population with $[3,3]$ morphology and nonnegative inputs. The direct-I rows have no explicit inhibitory neurons; the input stream drives learned branch-level inhibitory conductance banks directly. The explicit-I rows use a matched inhibitory population with $[3,3]$ morphology. `local_ca` updates explicit inhibitory-cell dendrites with the same policy used for excitatory dendrites, while `freeze` holds those dendrites fixed but still lets excitatory I-to-E synapses learn. Exact path transport closes the direct-I rank-1 gap, and explicit inhibitory cells nearly match both direct-I path transport and the explicit-I backprop ceiling.

Cue feedback mode	Context decode	MI	Corr(I,R)
rank-1	0.963 ± 0.032	0.867 ± 0.100 bits	-0.608 ± 0.085
rank-2	0.856 ± 0.180	0.810 ± 0.213 bits	-0.602 ± 0.041
pathway-vector	0.852 ± 0.170	0.694 ± 0.300 bits	-0.627 ± 0.028
path transport	0.816 ± 0.153	0.692 ± 0.227 bits	-0.642 ± 0.018

Table S9: **Fitted direct-I cue-routing diagnostic (5 seeds per learned-I LocalCA mode).** In `input_mode=1`, inhibitory synapse banks receive a nonnegative input stream directly. In trained cue-routing models, the balance of inhibitory activation across routed pathways carries context information and higher inhibitory conductance coincides with lower local input resistance. Context decoding is reported with best polarity because branch sign can flip across seeds; it should be read as evidence for context-dependent inhibitory path gating, not as a fixed anatomical assignment of one branch to one cue.

checkpoints while replacing the branch-level inhibitory conductance with four post-training interventions: zero inhibition, sample-shuffled inhibition, per-branch batch means, or one uniform matched mean. On a 512-example held-out subset, learned MNIST inhibition at N_I=5 gives $90.4 \pm 1.1\%$ accuracy, while the four interventions drop to 19.8–26.4%. On noise resilience at N_I=10, learned inhibition gives $80.5 \pm 2.4\%$, while the interventions drop to 10.3–19.3%. Thus the inhibitory field is not interchangeable with a matched global conductance load. We also computed the SVD of exact compartment-error matrices across samples and compartments. At matched MNIST N_I=5, shunting has lower rank-1 residual than additive (0.65 ± 0.18 vs. 0.83 ± 0.07) and lower participation rank (3.2 ± 1.2 vs. 8.5 ± 3.3), supporting the path-gain compressibility interpretation.

We also computed the direct log-gain covariance margin in Eq. 8 on the selected morphology diagnostic checkpoints. The margin is zero when no inhibitory conductance is present and slightly negative in the inhibited shunting selections (-2.5×10^{-3}, -5.0×10^{-4}, and -2.8×10^{-4} at N_I=5,20,40). We therefore do not use this covariance condition as a positive headline result. Its role is to clarify when shunting should compress gains; the empirical claim rests on the measured comparative path-gain dispersion, exact-error rank, and inhibition-intervention diagnostics.

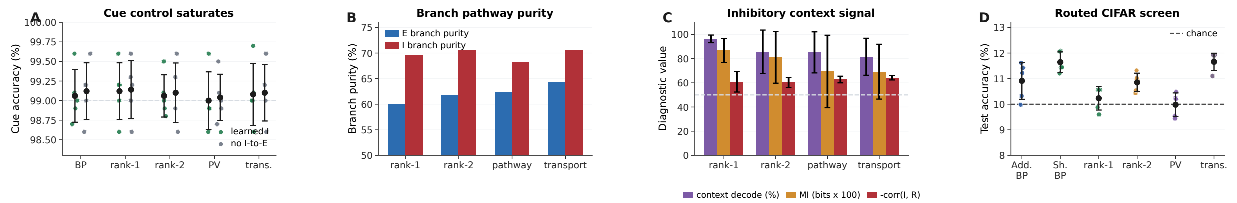

Figure S14: **Positive routed-task pathway and inhibition controls.** **(A)** Strictly nonnegative cue-routing controls saturate near 99% for learned-I and no-I variants, so this rerun is not a performance separation. **(B)** Fitted excitatory and inhibitory synapse banks retain branch-pathway structure above the 50% chance level. **(C)** In learned-I cue models, pathway-inhibition balance carries context information and inhibitory conductance is negatively coupled to local input resistance. **(D)** A nonnegative Double-CIFAR contextual screen remains near chance even under matched backprop, so it is not used as evidence for the paper's local-learning claims. Error bars show s.d. across 5 seeds.

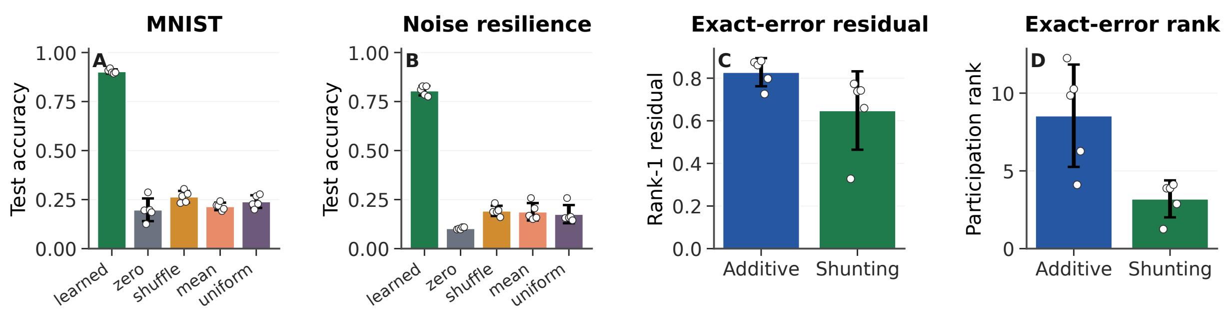

Figure S15: **Causal inhibition and exact-error compressibility.** **(A,B)** Post-training inhibitory-conductance interventions on trained shunting LocalCA checkpoints. Accuracy is measured on a 512-example held-out subset; dots show seeds. Removing, shuffling, or clamping learned inhibition collapses performance, showing that the learned inhibitory field carries sample-specific task structure. **(C,D)** SVD diagnostics of the exact compartment-error matrix on matched MNIST N_I=5 runs. Shunting has a smaller rank-1 residual and lower participation rank than additive, so the exact error field is more compressible. Error bars are s.d. across 5 seeds.

Cue-Routing Feedback-Rank Diagnostic

Cue routing tests the regime where one shared teaching signal is expected to fail: context determines which noisy cue is reliable, so different pathways should receive different credit. Figure S16 shows that the benchmark is learnable and that moving beyond rank-1 feedback helps, but the current path-structured variant does not yet beat the best unstructured low-rank setting. We therefore interpret this result as a failure-mode and rank-requirement diagnostic, not as a solved structured-feedback method.

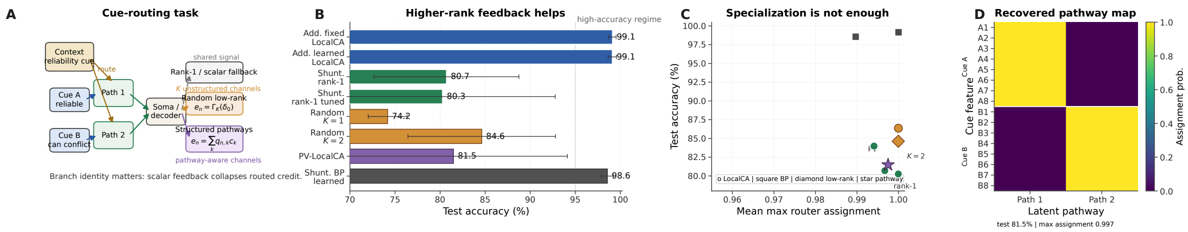

Figure S16: **Structured feedback when rank-1 broadcast fails. (A)** Cue-routing task schematic. Context determines which noisy cue is reliable on each trial, so different dendritic pathways should receive different teaching signals. **(B)** Five-seed comparison of rank-1, random low-rank, and pathway-structured feedback. Moving beyond rank-1 helps, while the best structured repair remains open. **(C)** Test accuracy vs. router specialization across learned-router conditions. High specialization alone does not guarantee success. **(D)** Recovered pathway assignments for a representative learned-router run. Cue-A and cue-B features segregate into separate latent pathways even when feedback quality, rather than router discreteness, is the limiting factor.

Setting	Broadcast	Test accuracy
Noise resilience, shunting, rank bridge		
per-soma	rank-1 per-soma	0.662 ± 0.016
path propagation	recursive attenuation proxy	0.762 ± 0.047
low-rank	$K{=}1$	0.453 ± 0.126
low-rank	$K{=}2$	0.630 ± 0.042
low-rank	$K{=}4$	0.764 ± 0.025
low-rank	$K{=}8$	0.795 ± 0.019
path transport	exact conductance-stage transport	$\mathbf{0.847 \pm 0.018}$
Cue routing, learned shunting router, rank/structure sweep		
per-soma	tuned rank-1	0.803 ± 0.125
low-rank	$K{=}1$	0.742 ± 0.022
low-rank	$K{=}2$	$\mathbf{0.846 \pm 0.082}$
low-rank	$K{=}4$	0.748 ± 0.085
pathway-vector	tuned structured rank-2	0.815 ± 0.126

Table S10: **Rank bridge and rank-structure comparison.** On shunting noise resilience, both the current path-propagation proxy and higher-rank random low-rank broadcast close a substantial fraction of the gap between rank-1 per-soma and exact path transport, with low-rank continuing to improve up to $K{=}8$. On cue routing, the learned-router sweep shows that moving beyond rank-1 helps, but the best current unstructured low-rank setting ($K{=}2$) is still slightly better than the tuned structured pathway-vector variant. We therefore interpret routed credit assignment as an open higher-rank problem rather than as a solved pathway-vector rescue.

Random Low-Rank Broadcast Channels

Frozen Activation-Choice Audit

B Implementation Details

Biological Plausibility Assumptions

The theorems and rules above rest on a small set of explicit assumptions. We state them as a checklist so that readers can see exactly what is assumed and where each assumption is used:

Dataset / strategy	Core	N_I	**Identity** (`none`)	**Frozen** `param_tanh`
MNIST / LocalCA	additive	0	0.780 ± 0.014	0.709 ± 0.037
MNIST / LocalCA	additive	10	0.907 ± 0.003	0.881 ± 0.005
MNIST / LocalCA	shunting	0	0.841 ± 0.007	0.778 ± 0.020
MNIST / LocalCA	shunting	10	0.905 ± 0.009	0.840 ± 0.016
Noise resilience / LocalCA	additive	0	0.590 ± 0.162	0.109 ± 0.006
Noise resilience / LocalCA	additive	10	0.859 ± 0.001	0.772 ± 0.009
Noise resilience / LocalCA	shunting	0	0.411 ± 0.028	0.331 ± 0.020
Noise resilience / LocalCA	shunting	10	0.769 ± 0.004	0.701 ± 0.006
MNIST / backprop	additive	0	0.925 ± 0.001	0.942 ± 0.001
MNIST / backprop	additive	10	0.921 ± 0.001	0.950 ± 0.001
MNIST / backprop	shunting	0	0.963 ± 0.0004	0.950 ± 0.001
MNIST / backprop	shunting	10	0.967 ± 0.001	0.946 ± 0.001
Noise resilience / backprop	additive	0	0.896 ± 0.003	0.106 ± 0.007
Noise resilience / backprop	additive	10	0.893 ± 0.002	0.927 ± 0.002
Noise resilience / backprop	shunting	0	0.898 ± 0.001	0.869 ± 0.002
Noise resilience / backprop	shunting	10	0.926 ± 0.001	0.912 ± 0.001

Table S11: **Frozen activation-shape audit on matched dendritic architectures (5 seeds per condition).** Identity (`none`) and a frozen bounded post-voltage reactivation `param_tanh` lead to markedly different operating regimes under nonnegative first-layer synaptic drive. In this audit, identity is the safer LocalCA choice on both MNIST and the noise-resilience task, while the backprop effect is strongly regime-dependent: frozen `param_tanh` helps additive most clearly in the inhibited noise-resilience setting (N_I=10) but collapses it at N_I=0. These runs match additive reactivation initialization to the shunting setting and hold the activation parameters fixed, isolating activation shape from both additive/shunting initialization differences and activation learning itself.

A1 (Steady-state voltage). We use the steady-state solution of the passive cable equation (Eq. 1). Temporal dynamics are not modeled; all gradients, path gains, and fidelity measurements refer to the conductance-stage voltage at steady state.

A2 (Tree topology). Each dendritic cell is a rooted tree with the soma as the root. There are no recurrent connections within the dendrite, which makes the path from any compartment to the soma unique and is what allows the closed-form path gain α_n in Theorem 1.

A3 (Nonnegative conductances). Synaptic and dendritic conductance parameters are pushed through a positive transform (`softplus`), so $g_j^{\text{syn}}, g_j^{\text{den}} \geq 0$ at every optimisation step. This is a biophysical constraint and is also what keeps the shunting denominator bounded away from zero.

A4 (Nonnegative presynaptic drive). The first synaptic drive x is nonnegative, implemented either by nonnegative inputs or an explicit ReLU transfer stage. Together with A3 this enforces the positive-input condition used for all headline sweeps.

A5 (Pre-reactivation upward transfer). Theorem 1 is stated for the pre-reactivation compartment voltage. A post-voltage reactivation f (e.g. `param_tanh`) is handled by multiplying the path gain by $f'(V_n)$ along the recursion. The compartment error remains the sole non-local quantity; only the effective path gain acquires local activation-derivative factors.

A6 (Single perisomatic broadcast). Every synapse receives one non-local scalar or low-rank vector signal

Dataset	Core	N_I	**Identity** (`none`)	**Frozen** `param_tanh`
MNIST	additive	0	CV 0.280, cosine 0.220	CV 1.122, cosine 0.074
MNIST	additive	10	CV 0.310, cosine 0.212	CV 1.079, cosine 0.094
MNIST	shunting	0	CV 0.393, cosine 0.170	CV 0.770, cosine 0.114
MNIST	shunting	10	CV 0.385, cosine 0.281	CV 1.177, cosine 0.106
Noise resilience	additive	0	CV 0.124, cosine 0.079	CV 0.114, cosine 0.052
Noise resilience	additive	10	CV 0.414, cosine 0.206	CV 0.840, cosine 0.094
Noise resilience	shunting	0	CV 0.255, cosine 0.324	CV 0.344, cosine 0.341
Noise resilience	shunting	10	CV 0.762, cosine 0.140	CV 2.497, cosine 0.079

Table S12: **Frozen activation-shape theory audit on matched LocalCA runs (5 seeds per condition).** Each entry reports conductance-stage path-gain dispersion (CV) and direct per-soma compartment-error fidelity (cosine between the per-soma broadcast and the exact pre-reactivation compartment error). The positive-input theory audit largely agrees with the empirical one: freezing `param_tanh` usually broadens path-gain dispersion and weakens per-soma fidelity, especially in the inhibited settings where the main mechanistic story lives. The one mild exception is low-inhibition shunting on noise resilience, where frozen `param_tanh` leaves fidelity roughly unchanged while still hurting final accuracy, suggesting that activation choice affects optimization and representation as well as direct credit fidelity.

delivered from the soma; all other ingredients of the update rule (eligibility traces, driving force, input resistance, covariance modulators) are computable from quantities available at the synapse or compartment.

Each synapse therefore has access to: presynaptic activity x_j (local), compartment voltage V_n (local membrane potential), reversal potential E_j (fixed by receptor type), and total input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$ (computable from local conductances). The 4F modulator ρ_n requires online estimation of voltage covariance (biologically plausible via slow calcium signals); the 5F factor ϕ_n requires a linear regression proxy (implementable via eligibility traces). The only non-local quantity is the broadcast error e_n, which requires a top-down or neuromodulatory signal from the soma to dendritic compartments. We assume per-neuron resolution (one error per output neuron) for the per-soma broadcast mode.

Dendritic Structure in Code

We use the morphology notation $[b_1, b_2, \ldots, b_D]$ for rooted dendritic trees, where each factor gives the fan-in from one dendritic stage to the next on the path to the soma. For example, $[3,3]$ means that each soma aggregates 3 proximal branches and each proximal branch aggregates 3 distal branches, for 9 distal leaves per soma; $[3,3,3]$ adds a third dendritic stage with 27 leaves per soma. In code, each excitatory unit is instantiated as a `DendriNet`, which is a stack of `DendriticBranchLayers`. Each branch layer contains an excitatory synapse bank, an optional inhibitory synapse bank, a branch-to-parent aggregation map, and an optional post-voltage reactivation. In the main feedforward models studied here, `somatic_synapses=false`, so synaptic inputs terminate on dendritic branches rather than directly on the soma. The headline runs use `input_mode=1`: the transfer layer creates nonnegative excitatory and inhibitory streams from the same external input, and nonzero `ie_synapses_per_branch_per_layer` attaches learned inhibitory conductance banks directly to excitatory dendritic branches. These models therefore have input-driven branch inhibition but no separate inhibitory neuron population. By contrast, `input_mode=0` instantiates explicit feedforward inhibitory cells when `inhibitory_layer_sizes` is nonempty; these cells can transform excitatory activity before projecting inhibitory conductance back onto excitatory dendrites. In that setting the inhibitory cells are also dendritic modules, so their E-to-I and I-to-I synaptic conductances are trained by the same local rule family rather

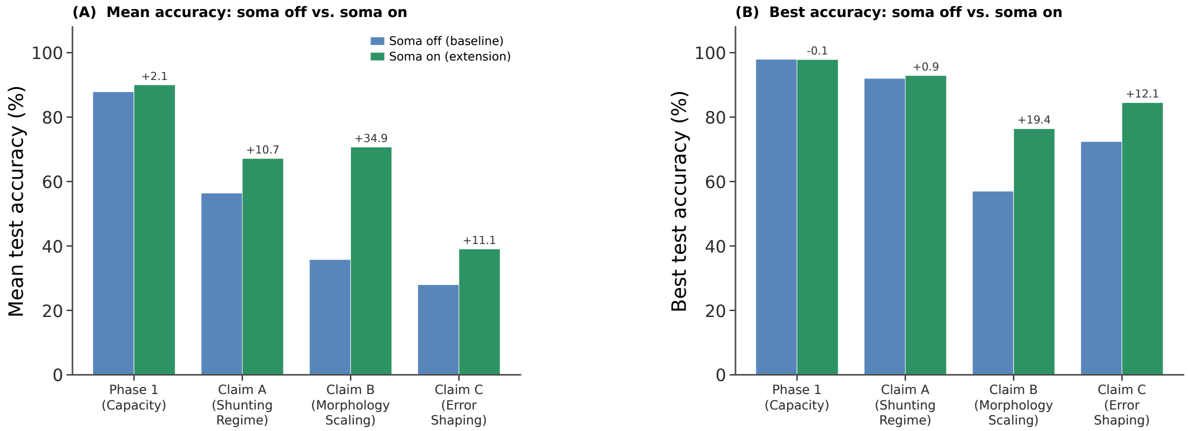

Figure S17: **Soma-on extension for LocalCA.** Enabling direct somatic input (`somatic_synapses=true`) improves aggregate mean test accuracy across all four manuscript LocalCA families. Each family is summarized over the completed configurations in matched soma-off and soma-on sweeps. The largest gain appears in morphology scaling, where the family mean rises from 35.8% to 70.7% and the best run rises from 57.0% to 76.4%. Gains in the shunting-regime and error-shaping families are smaller but still substantial. Delta labels show soma-on minus soma-off. Because the main mechanistic analysis was developed in the dendrite-only feedforward regime, we report soma-on as a supportive extension rather than replacing the main operating regime.

than by a separate global inhibitory optimizer. We treat that explicit-I setting as an exploratory mechanism probe rather than a headline baseline, because the main claims were measured in the direct-I-stream regime. The additive and shunting cores are architecture-matched at the tree level; they differ in the branch-level voltage integration rule rather than in whether a dendritic tree exists. In all reported dendritic runs, synaptic and dendritic conductance parameters are pushed through positive transforms (`softplus` in the main paper runs), so the conductances themselves remain nonnegative at every optimization step; unconstrained decoder weights are separate readout parameters and are not interpreted as conductances.

Matched soma-on LocalCA sweeps suggest that direct somatic input is a meaningful extension rather than a minor implementation detail. Relative to matched non-somatic sweeps, enabling `somatic_synapses=true` raises the mean test accuracy in all four manuscript families: phase-1 capacity from 0.8784 to 0.8999, shunting-regime sweeps from 0.5641 to 0.6715, morphology-scaling sweeps from 0.3578 to 0.7070, and error-shaping sweeps from 0.2798 to 0.3906. These numbers are family-level aggregates over the completed configurations in each matched soma-off and soma-on sweep, rather than a single 5-seed cell. Because the main mechanistic analysis in this paper was developed in the dendrite-only feedforward regime, we keep the main text anchored to `somatic_synapses=false` and treat soma-on as a supportive extension rather than silently replacing the original operating regime. Figure S17 summarizes the mean and best-accuracy comparisons across the four families.

The soma-on benefit extends beyond the capacity/scaling families to the cue-routing benchmark that stress-tests rank-1 feedback. With dendrite-only pathways, the best learned-routing shunting LocalCA configuration was unstable at 0.81 ± 0.13; with `somatic_synapses=true`, fixed-router shunting LocalCA reaches 1.00 ± 0.0 test accuracy across 5 seeds, and learned-router additive LocalCA reaches 0.99 ± 0.003 (Fig. S18). The soma channel provides a high-bandwidth direct input that bypasses the rank-1 bottleneck induced by per-soma broadcast.

Family	**Soma-off**	**Soma-on**	**Δ mean**	**Δ best**
Phase-1 capacity	0.8784	0.8999	+0.0215	-0.0007
Shunting regime	0.5641	0.6715	+0.1074	+0.0090
Morphology scaling	0.3578	0.7070	+0.3492	+0.1940
Error shaping	0.2798	0.3906	+0.1109	+0.1210

Table S13: **Representative soma-on deltas across the four LocalCA families.** Values compare matched families with and without `somatic_synapses` and report both the change in family-mean test accuracy and the change in best-run test accuracy. Cue-routing soma and CIFAR-10 depth-4 soma are reported separately in Fig. S18, Fig. S10, and Table S7 because those extensions are summarized on different task families.

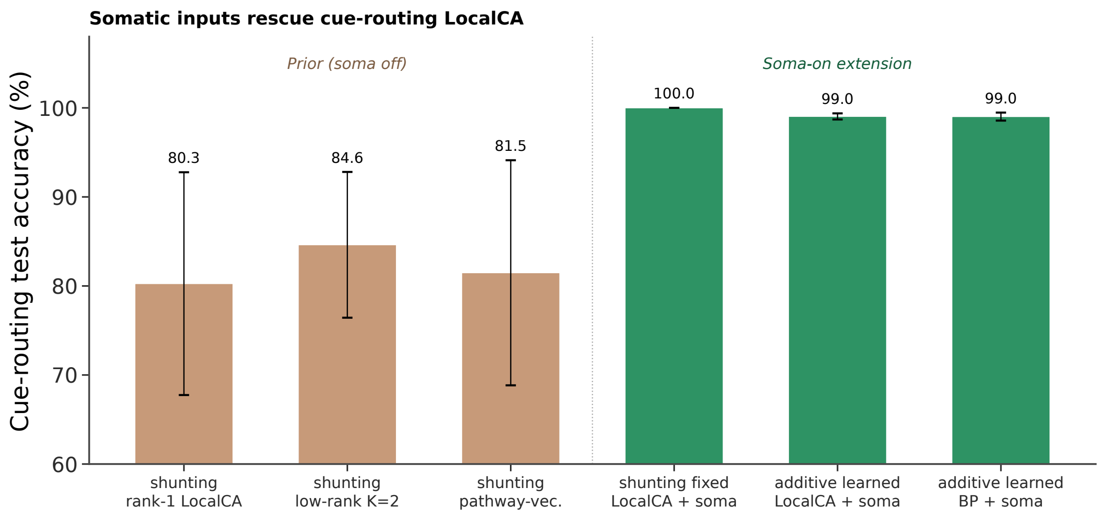

Figure S18: **Soma-on rescues cue-routing LocalCA.** The left bars show the previously reported soma-off LocalCA results on the cue-integration benchmark. The right bars show the soma-on extension: fixed-router shunting LocalCA reaches 1.00 test accuracy, and learned-router additive LocalCA with soma reaches 0.99, matching the additive backprop ceiling with soma. The soma channel supplies a direct high-bandwidth input that bypasses the rank-1 bottleneck identified in the main text. Learned-router shunting with soma is omitted because this benchmark uses signed cues, which conflict with the shunting positive-input check in the learned-router pipeline; fixed-router shunting is therefore the clean shunting data point.

Post-Voltage Reactivation

The conductance stage produces a pre-reactivation compartment voltage V_n, after which each branch layer may apply a monotone scalar nonlinearity. The implementation supports identity (`none`), fixed `tanh`/`sigmoid`/`relu`, and several parametric families including `param_tanh`, `param_tanh_only_m`, and piecewise linear-saturating variants. The main manuscript sweeps use `param_tanh`, a learnable bounded firing-rate transform of the form $(\tanh(m(V-b))+1)/2$, together with an explicitly configured nonnegative transfer stage so that the first synaptic drive remains nonnegative even after input corruption or structured task noise. Most headline feedforward sweeps use a linear decoder, while the CIFAR extension uses a nonlinear decoder and therefore maps output error back to decoder-input/soma space through the decoder Jacobian before applying LocalCA. Unless explicitly overridden, additive and shunting cores can start from different reactivation initializations; activation-fairness ablations therefore match additive initialization to the shunting setting when comparing alternative reactivation choices directly. The appendix activation audit additionally fixes `param_tanh` after initialization to isolate activation shape from activation plasticity.

Quantity	Symbol	Convention
Voltage	V	Normalized to $[-1,1]$
Conductances	$g^{\text{syn}}, g^{\text{den}}$	Nonneg. via softplus
Leak conductance	g^{leak}	Set to 1
Input resistance	R^{tot}	≤ 1

Table S14: Units and normalization.

LocalCA uses occupancy-quantile initialization together with a calibration pass that floors tiny quantile widths, clamps extreme fitted slopes, and reverts invalid per-layer fits rather than allowing pathological intermediate values to propagate. In the non-somatic feedforward publication families, this calibration leaves the results essentially unchanged relative to the same occupancy-quantile setup without the additional validity checks: phase-1 capacity changes from 0.8786 to 0.8784 mean test accuracy, shunting-regime sweeps from 0.5617 to 0.5641, morphology-scaling sweeps from 0.3600 to 0.3578, and error-shaping sweeps from 0.2860 to 0.2798. The calibration therefore improves numerical safety without changing the qualitative LocalCA story.

We also compared two post-initialization policies for the reactivation parameters under quantile initialization: *learned local* (b,m), where LocalCA updates the gate parameters through the local rule, and *quantile-maintained* (b,m), where (b,m) are periodically refreshed from the current voltage distribution instead of being updated by the local rule. This comparison matters because quantile-maintained gates substantially simplify nonlinear reactivation learning: the rest of the network still uses LocalCA, but the gate parameters themselves are set by a local distributional rule rather than a learned plasticity rule. On MNIST classification sweeps, learned local (b,m) remains clearly better (additive 0.9247 vs. 0.9061 mean test accuracy; shunting 0.9104 vs. 0.8766). On CIFAR-10, the picture is mixed in a useful way: additive is essentially tied on mean (0.4223 learned-local vs. 0.4215 quantile-maintained) while shunting improves under quantile-maintained gates (0.4028 to 0.4356 mean test accuracy). These bars aggregate over a matched morphology/init grid rather than a dedicated 5-seed rerun, so we treat them as a supportive policy comparison rather than a headline result. Accordingly, the main manuscript continues to use occupancy-quantile initialization followed by LocalCA updates of (b,m) as the default local-learning setting, while quantile-maintained (b,m) is reported as a supplementary alternative. Figure S19 shows the full MNIST and CIFAR-10 comparison.

Component Ablation

To isolate the contribution of each ingredient in our LocalCA recipe, we ran a per-component ablation on MNIST starting from the manuscript default and removing one component at a time. The default combines quantile init, learned local (b,m), `somatic_synapses=false`, and `param_tanh` reactivation. We also include `somatic_synapses=true` as a positive extension and matched standard backpropagation as the upper reference. Figure S20 reports test accuracy (5 seeds per cell) for both shunting and additive dendritic cores on clean MNIST with morphology $[3,3]$.

Units and Parameterization

Decoder Update Modes

W_{dec} maps $V_L \to \hat{y}$. Modes: **backprop** ($\nabla_W L$ via autograd), **local** ($\Delta W = \eta \langle \delta_0 V_L^\top \rangle_B$), **frozen** ($\Delta W = 0$).

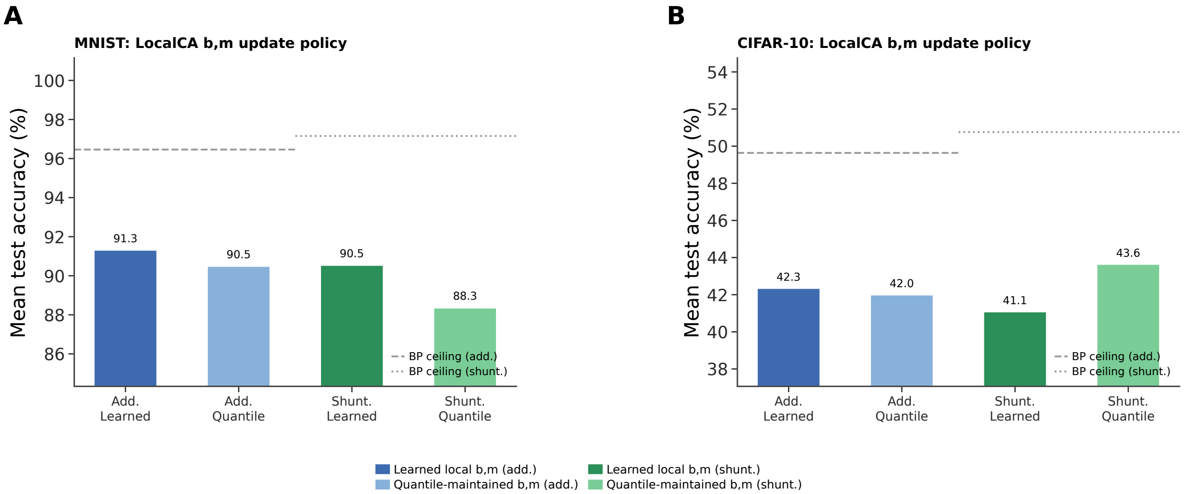

Figure S19: (b,m) **update policy under quantile initialization.** Each cluster of bars compares the mean test accuracy of learned local (b,m) and quantile-maintained (b,m) under LocalCA for additive and shunting cores, averaged across a matched morphology/init grid ($[2,2]$ and $[3,3,3]$; analytical and quantile init variants). Dashed and dotted horizontal lines mark the matched standard backpropagation ceiling for additive and shunting respectively. **Left (MNIST):** Learned local (b,m) is consistently better; quantile-maintained lags by 1.9 pp (additive) and 3.4 pp (shunting) on mean accuracy. **Right (CIFAR-10):** The gap reverses for shunting: quantile-maintained reaches 43.6% mean vs. 40.3% for learned local, while additive is essentially tied. The pattern suggests that quantile-maintained (b,m) prevents over-fitting the reactivation gate on harder inputs, making it a useful alternative in harder visual regimes.

Category	Implemented modes	Role in this paper
Core models	point MLP; dendritic MLP; dendritic additive; dendritic shunting; pathway-routed dendritic cores	Main comparisons use dendritic additive vs. dendritic shunting; routed cores are used for cue integration.
Training strategies	standard backprop; LocalCA; DFA; FA; transported-broadcast oracle	Main text uses standard and LocalCA; DFA/FA remain appendix baselines; path transport is an oracle upper bound.
Broadcast modes	scalar; per-soma; local mismatch; low-rank; path transport; pathway vector	Main text centers on per-soma, low-rank, and transported-error controls; scalar and local-mismatch are supporting controls, while pathway vector remains an appendix structured-feedback extension.
Recent dendritic extensions	path propagation; router-derived branch roles; pathway-vector feedback; inhibitory homeostasis	These were added to test morphology-aware credit transport and structured branch-specific plasticity beyond the original 3F/4F/5F family.

Table S15: Implemented model families and LocalCA operating modes. Only a subset is used for the main claims; the remaining modes serve as controls, ablations, or structured extensions.

Model and Mode Inventory

Representative Manuscript Architectures

Hyperparameters

Table S17 collects the core training hyperparameters for each headline experiment. Learning rates are applied through parameter groups (top-k, dendritic block-linear, reactivation, decoder) with the values given in the

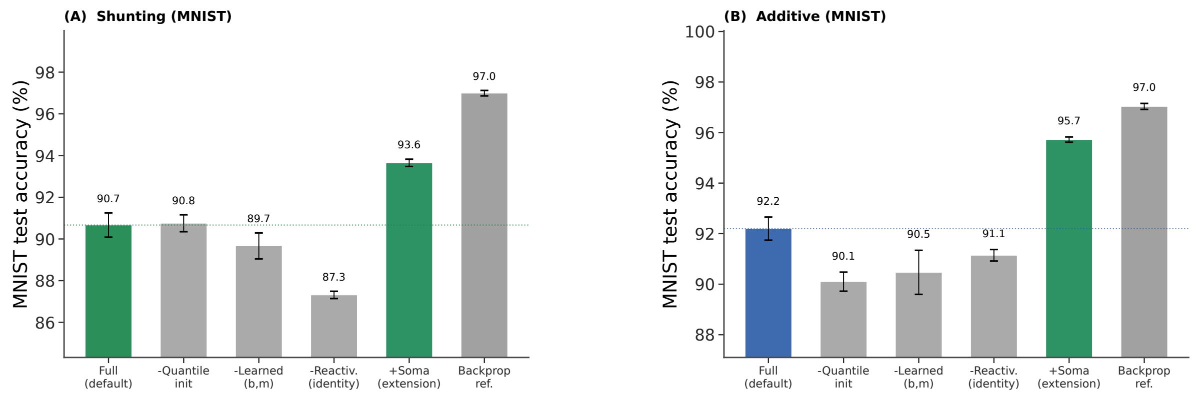

Figure S20: **Per-component ablation for LocalCA on MNIST.** Bars show mean test accuracy ($\pm$ s.d. across 5 seeds); coloured bars highlight the manuscript default (blue/green) and the soma-on extension (green). **(A) Shunting:** The full default reaches 90.7%; removing quantile init leaves accuracy unchanged at 90.8% (quantile init is a mild calibration nudge on this clean task), removing learned (b,m) drops by about 1 pp, and removing reactivation entirely drops accuracy to 87.3%; the reactivation transform is the largest single LocalCA ingredient. **(B) Additive:** Quantile init contributes $+2.1$ pp ($90.1 \to 92.2\%$), learned (b,m) contributes $+1.7$ pp, and reactivation contributes $+1.1$ pp; each ingredient helps on the additive core. **Soma-on** is the largest positive change in both cores ($+3$ to $+3.5$ pp over full config), while matched standard backpropagation reaches 97.0% in both cores and sets the upper reference. The fixed-`relu` reactivation condition did not satisfy the same initialization checks and is omitted.

representative config files; where a single LR is reported, every group uses that value.

Effective Broadcast Dimensionality in Headline Architectures

LocalCA Extension Compatibility

Seed Protocol and Evidence Map

The seed depth is not uniform across every sweep, and we therefore separate main evidence from supporting diagnostics. Table S20 pins each figure or table in this paper to its seed count and evidence tier.

Compute Resources

All experiments were run on an internal GPU cluster. Most MNIST, Fashion-MNIST, and synthetic-task runs used a single NVIDIA A100 or H100 GPU and finished in minutes to tens of minutes per seed. The flattened CIFAR-10 runs required longer single-GPU jobs but still did not use distributed or multi-GPU training. Reported headline metrics are aggregated over 5 seeds unless noted otherwise. The remaining 3-seed results are confined to selected supportive sweeps, and the broader exploratory screens used to tune and diagnose the method required additional compute beyond the final reported runs.

Code, Data, and Asset Availability

MNIST, Fashion-MNIST, CIFAR-10, and MICrONS are public datasets or public benchmark assets cited in the paper; we use them under their standard published access conditions and cite their original sources in the bibliography. The present anonymous submission does not include a public code URL in order to preserve double-blind review, but the paper specifies the model family, training modes, key hyperparameters, and representative configurations, and we plan to release code and configuration files after review. The paper does not release a new dataset or a stand-alone pretrained model asset.

Algorithm

Algorithm 1 Local Credit Assignment

Input: Model, batch (x, y), config $\mathscr{C}$
Forward pass; loss L, output error δ^y
Somatic/core error $\delta_0 =$ decoder-input Jacobian product $J_{\text{dec}}(h_{\text{core}})^\top \delta^y$ (or $W_{\text{dec}}^\top \delta^y$ for a linear decoder)
for each layer n (reverse) **do**
    $e_n = \text{broadcast}(\delta_0, \mathscr{C})$; optionally gate by pathway roles
    Compute ρ_n, ϕ_n (EMA estimators, if enabled)
    Apply the configured local update (Eq. 10 or 11)
end for
Clip gradients; optimizer step

C Theoretical Details

Variant Taxonomy

Random-Broadcast Alignment

Proposition 4 (Supportive random-broadcast alignment)**.** *Let the broadcast matrix B_n have i.i.d. zero-mean entries with $\mathbb{E}[B_n^\top B_n] = \alpha I$. Then the expected cosine between the resulting local gradient and the exact gradient is positive, $\mathbb{E}[\cos\angle(g^{\text{local}}, g^{\text{exact}})] \geq c_n > 0$, by an argument analogous to feedback alignment [9]. The constant c_n depends on how strongly local eligibility factors correlate with the conductance-stage path gain in Eq.* (4)*; shunting improves this correlation by narrowing the spread of intermediate sensitivities.*

We treat this argument as supportive rather than central. The main theoretical object in the paper is still the exact pre-reactivation compartment error $\partial L/\partial V_n$; under identity upward transfer this reduces to $\alpha_n \delta_0$, while with post-voltage activation the corresponding effective path gain additionally includes local $f_n'(V_n)$ factors. The main empirical question is how faithfully different broadcast fields approximate that quantity.

Biological Analogs

D Morphology-Aware Extensions

Path-integrated propagation. Modulate broadcast error by $\pi_n = \pi_{n-1} \cdot R_{n-1}^{\text{tot}} \cdot \bar{g}_{n-1}^{\text{den}}$, approximating depth attenuation from Eq. 4.

Depth modulation. Per-branch scaling $\rho_j = \rho_{\text{base}}/(d_j + \alpha)$, mirroring cable attenuation.

Dendritic normalization. $\Delta g_j^{\text{den}} \leftarrow \Delta g_j^{\text{den}}/(\sum_k g_k^{\text{den}} + \varepsilon)$, analogous to homeostatic scaling [21].

Pathway-vector feedback. For tasks with latent pathway structure, the broadcast becomes a role vector inferred from the router. Local pathway activity gates this vector. Upward block transport then passes it to earlier branches, aligning their feedback with downstream descendants.

Router-derived branch roles. Each branch receives a role profile inferred from router assignments and incoming block weights rather than from a hand-coded apical/basal label. Plasticity is then modulated by branch selectivity and a synapse-specific role-alignment factor, emphasizing pathway-consistent updates without imposing a heuristic branch taxonomy.

E HSIC Auxiliary Objectives

Following [17], we use kernel-based HSIC objectives:

$$\mathscr{L}^{\text{self}} = B^{-2}\,\mathrm{tr}(\mathbf{K}_Z\mathbf{H}\mathbf{K}_Z\mathbf{H}), \qquad \mathscr{L}^{\text{target}} = -B^{-2}\,\mathrm{tr}(\mathbf{K}_Z\mathbf{H}\mathbf{K}_Y\mathbf{H}).$$

Moderate weights (0.01–0.1) help on context gating; negligible on MNIST. Online statistics (ρ_n, ϕ_n) use Welford's algorithm [20].

F Online Variant with Eligibility Traces

The continuous-time eligibility trace is

$$\tau_e \dot{e}_j^{\text{syn}} = -e_j^{\text{syn}} + x_j(E_j - V_n)R_n^{\text{tot}},$$

with update

$$\Delta g_j^{\text{syn}} \propto \int e_j^{\text{syn}}(t)m_n(t)\,\mathrm{d}t$$

[18, 19].

G Weight Statistics and Biological References

As a secondary analysis, we examine whether the conductance-based update naturally induces characteristic weight statistics. Because synaptic updates scale with input resistance $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$, the voltage equation implies multiplicative weight dynamics, creating a regime analogous to exponentiated-gradient updates. Such multiplicative dynamics suggest log-normal weight distributions in trained networks [38, 39]. We compare the resulting distributions against two biological references: electrophysiological data from Song et al. [38] (rat V1 L5 pyramidal, $\sigma = 0.94$) and the MICrONS mm^3 connectome [40] (mouse V1 synapse sizes).

Model weight distributions are log-normal. We fit log-normal distributions to the active (top-K masked) excitatory synaptic weights after training on MNIST and Fashion-MNIST. Table S24 shows the fitted σ (standard deviation of $\log w$) for each condition. Shunting consistently produces narrower distributions than additive integration, and backpropagation produces narrower distributions than local learning. These two factors combine approximately additively: BP+Shunting $\rightarrow$ narrowest, Local+Additive $\rightarrow$ broadest.

Shunting is closest to the biological references considered here. BP+Shunting produces $\sigma = 0.94$ on MNIST, very close to the Song et al. reference ($\Delta = 0.008$). Local+Shunting produces $\sigma = 1.13$–1.22, close to MICrONS E$\rightarrow$E synapse sizes ($\sigma = 1.14$; $\Delta \leq 0.08$). By contrast, Local+Additive produces $\sigma = 1.43$–1.69, broader than any biological reference considered. This pattern is consistent across datasets (MNIST and Fashion-MNIST), suggesting it reflects structural properties of the learning rule rather than task-specific features.

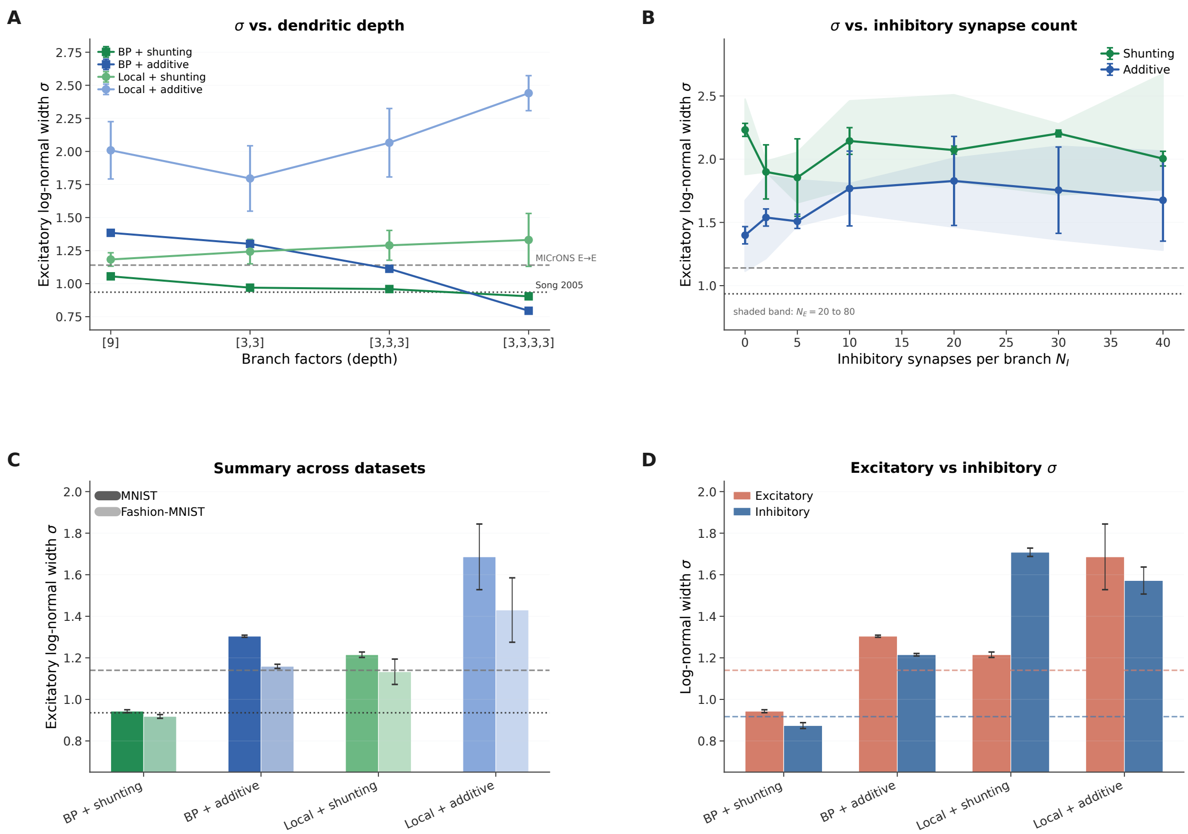

Figure S21: **Synaptic weight distribution sensitivity to architecture.** **(A)** Log-normal σ vs. dendritic depth (branch factors [9] to [3,3,3,3]). Both backprop and local shunting remain closer to the biological-width references than their additive counterparts across depth. **(B)** σ vs. inhibitory synapses per branch (N_I). Without inhibition (N_I=0), all conditions are overly broad; with biologically constrained N_I=10–20, shunting enters the biological range. Shaded bands show the sweep over N_E=20 to 80. **(C)** Summary of MNIST and Fashion-MNIST results with biological references. The shunting ordering is consistent across datasets. **(D)** Excitatory vs. inhibitory weight σ on MNIST with MICrONS E→E and I→E reference bands.

The match depends on inhibitory conductance. We test sensitivity to the E/I synapse ratio using a sweep over $N_I \in \{0, 2, 5, 10, 20, 30, 40\}$ inhibitory synapses per branch (Appendix Fig. S21). Without inhibition (N_I=0), all conditions produce $\sigma > 2.0$, far from biology. With biologically constrained ratios (N_I=10–20, corresponding to E/I ratios of 4:1 to 2:1; [42]), shunting networks enter the biological range ($\sigma \approx 0.9$–1.2), while the effect of N_I on additive networks is less systematic. The weight distribution shape is also robust to dendritic depth (branch factors [9] through [3,3,3,3]; see Appendix Fig. S21A): BP+Shunting converges toward Song et al. with increasing depth, while Local+Shunting remains near MICrONS E→E across depths.

H Weight Distribution Sensitivity Analysis

Side-by-side BP vs. LocalCA across depths. To complement the per-architecture analysis above, we ran a matched BP vs. LocalCA sweep over three branch-factor settings ([2,2], [3,3], [3,3,3]) under both shunting and additive cores on MNIST (5 seeds per cell; quantile init; learned (b, m) for LocalCA). Figure S22 shows that at shallow and medium depths the local rule and backprop reach broadly similar excitatory weight means ($\approx$ 0.8–0.9), while the coefficient of variation (CV) under LocalCA is roughly twice as large as under BP

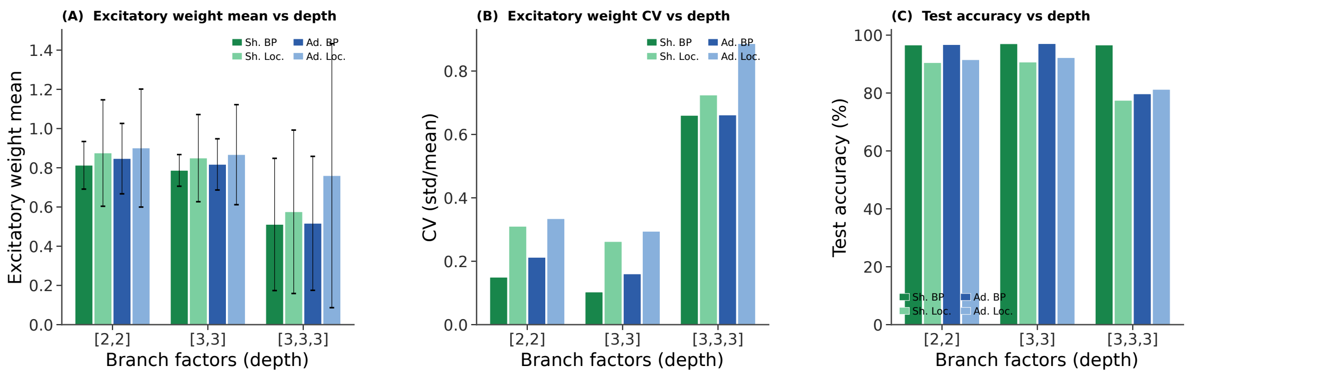

Figure S22: **Synaptic weight statistics vs. dendritic depth (MNIST, 5 seeds/cell). (A)** Excitatory weight mean (error bars = std). At $[2,2]$ and $[3,3]$, all four conditions cluster around 0.8–0.9; at $[3,3,3]$ the means drop to ≈ 0.5–0.75 and the cross-seed spread grows, especially for additive LocalCA. **(B)** Coefficient of variation (CV = std/mean). At shallow/medium depth, LocalCA produces $\sim 2\times$ higher CV than matched BP, but both remain well-behaved (≤ 0.33). At the deepest network all four conditions jump to CV ≈ 0.66–0.88, with additive LocalCA the broadest. **(C)** MNIST test accuracy. BP tracks $\sim$96–97% at $[2,2]$/$[3,3]$ and $\sim$92–95% at $[3,3,3]$. LocalCA stays within $\sim$5–6 pp of BP at shallow/medium depth but trails by $\sim$15–20 pp at $[3,3,3]$, identifying depth as the remaining open challenge for this local rule. The qualitative ordering (shunting LocalCA tracks shunting BP more closely than additive LocalCA tracks additive BP) reinforces the manuscript claim that the conductance denominator is the key mechanistic ingredient.

at these depths (e.g. shunting $[2,2]$: $\mathrm{CV}_{\mathrm{BP}} = 0.15$ vs. $\mathrm{CV}_{\mathrm{Loc.}} = 0.31$; shunting $[3,3]$: 0.11 vs. 0.26). At the deepest setting $[3,3,3]$ the weight means drop and the CV jumps substantially for both optimizers (CV ≈ 0.66–0.88), and test accuracy drops by $\sim$5 pp for BP and $\sim$15–20 pp for LocalCA. Across the explored depth range, shunting LocalCA stays closer to shunting BP than additive LocalCA does to additive BP on both weight and accuracy axes, consistent with the manuscript claim that the conductance denominator is the key mechanistic ingredient.

Weight distributions under soma-on. To check whether the direct somatic synapse extension (Fig. S17) alters the weight-distribution picture, we ran a matched 5-seed sweep at depth $[3,3]$ on MNIST with `somatic_synapses=true` across all four (core, strategy) cells; results are summarized in Fig. S23. Two effects emerge. First, shunting + BP becomes *tighter* than in the soma-off case: excitatory-weight CV drops from ≈ 0.11 (soma-off, $[3,3]$) to ≈ 0.10 (soma-on), consistent with the soma sink absorbing part of the drive and reducing per-branch variance. Second, shunting + LocalCA goes in the opposite direction: CV grows from ≈ 0.26 (soma-off) to ≈ 0.40 (soma-on), while MNIST test accuracy drops $\sim$3 pp (from 96.5% to 93.6%). Additive cores sit between these extremes (CV ≈ 0.23–0.30, accuracy 95.7–97.0%) and are only weakly affected. The ordering is consistent with the mechanism in the main text: the somatic synapse is an additional BP-visible path that backprop exploits cleanly, but under the local rule it adds broadcast variance that the scalar/per-soma signal cannot sharpen, widening the weight distribution without a matching accuracy gain.

MICrONS connectome analysis. We extracted synapse size distributions from the MICrONS mm^3 condensed connectome (v1181; 9.0M connections, 71.9K neurons; 89% excitatory). Splitting by pre/post cell type: E$\to$E connections (n=3.85M) show σ_{size}=1.14 and CV=1.01; I$\to$E (n=3.53M) show σ=0.92 and CV=1.03; E$\to$I (n=1.11M) show σ=0.83; I$\to$I (n=0.51M) show σ=0.85. Inhibitory connections are more frequently multi-synaptic (I$\to$E: 37.5% multi-synapse; E$\to$E: 11.5%), and inhibitory synapses are on average

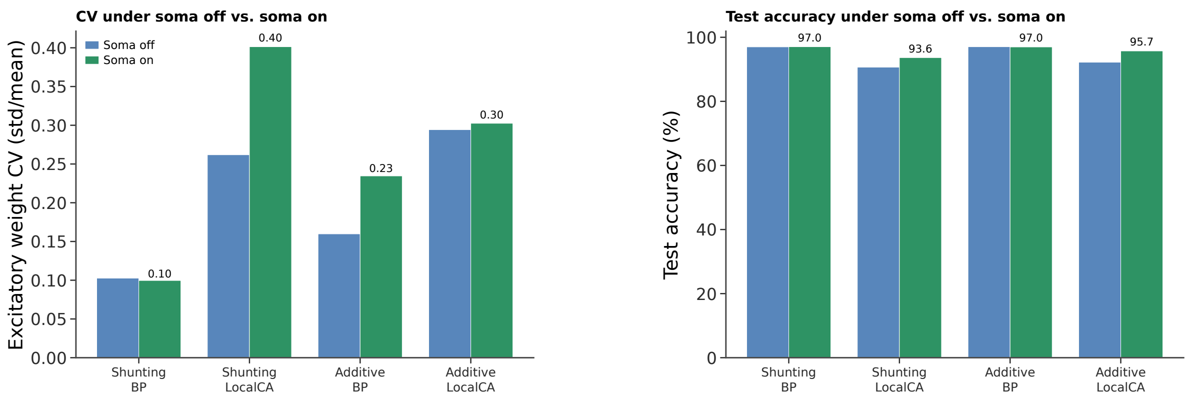

Figure S23: **Excitatory-weight CV and MNIST accuracy under soma-off vs. soma-on (depth** $[3,3]$**, 5 seeds/cell). (A)** Coefficient of variation of excitatory weights for the four (core, strategy) cells; paired bars show soma-off (from Fig. S22) and soma-on (this sweep). Shunting + BP tightens slightly; shunting + LocalCA broadens; additive cells change little. **(B)** Matched MNIST test accuracy (sanity panel). BP accuracy is preserved or marginally improved; LocalCA accuracy is preserved for additive but drops $\sim$3 pp for shunting, consistent with the CV widening in panel A.

smaller (median 3532 vs. 4676 for E→E).

Sensitivity to ee_synapses. Using the E/I grid sweep ($N_E \in \{20, 40, 80\}$ and $N_I \in \{0, 2, 5, 10, 20, 30, 40\}$; 168 models), excitatory σ increases modestly with N_E in both architectures, but the shunting-versus-additive ordering is preserved at every N_E value. The dominant driver of σ is still N_I: moving from N_I=0 to N_I=10 reduces shunting σ by roughly 1.0–1.5 units, while the effect on additive is smaller and less systematic.

Sensitivity to depth. From the depth-scaling sweep ($[9]$, $[3,3]$, $[3,3,3]$, $[3,3,3,3]$; 80 models, 5 seeds each), BP+Shunting excitatory σ decreases with depth ($1.06 \to 0.90$), converging toward Song et al. (0.94). Local+Shunting σ increases slightly ($1.18 \to 1.33$) but remains in the biological range at shallow depths. BP+Additive shows the strongest depth sensitivity ($1.38 \to 0.79$), while Local+Additive diverges with depth ($2.01 \to 2.44$). The morphology-aware modulators (path propagation, depth/centrality weighting) have negligible effect on σ.

Mechanistic interpretation. The shunting conductance $g_n^{\text{tot}} = g_n^{\text{leak}} + \sum g^{\text{exc}} + \sum g^{\text{inh}}$ appears in the update factor $R_n^{\text{tot}} = 1/g_n^{\text{tot}}$. As inhibitory conductance increases, g_n^{tot} grows and R_n^{tot} shrinks, compressing the multiplicative scale of weight updates. This is analogous to a learning rate that self-adjusts via the total conductance: strong synapses contribute more to g_n^{tot}, reducing their own future update magnitude. This acts as a natural form of weight-dependent regularization. The resulting equilibrium distribution is log-normal, consistent with Loewenstein et al. [41] and related multiplicative spine-dynamics analyses. Additive integration lacks this self-normalizing mechanism, producing broader distributions that fall outside the biological range.

I Synthetic Benchmark Descriptions

Each synthetic task is generated procedurally from a fixed random seed and uses fixed train/val/test splits for reproducibility. The MNIST-derived synthetic tasks use flattened 28×28 inputs; cue integration instead uses a small vector input with two output classes. Output dimensionality therefore varies by task and is stated in

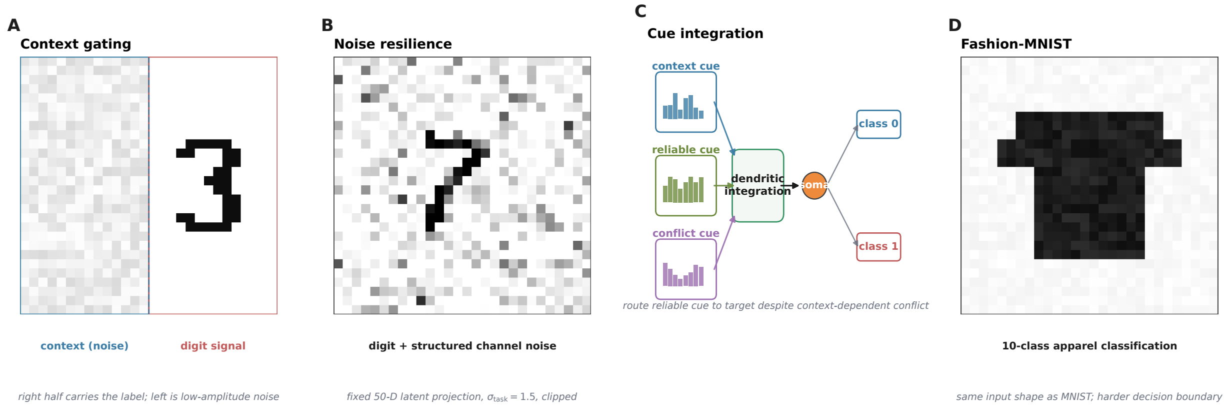

Figure S24: **Synthetic benchmark schematics. (A)** Context gating: the right half carries the MNIST digit; the left half is filled with low-amplitude uniform noise. **(B)** Noise resilience: MNIST image corrupted by structured channel noise (fixed 50-D latent projection, $\sigma_{\text{task}} = 1.5$, clipped to $[0,1]$). **(C)** Cue integration: a context cue selects which of two noisy cue streams is reliable; dendritic integration routes the reliable cue to the binary readout. **(D)** Fashion-MNIST: 10-class apparel classification sharing the MNIST input shape.

the task description when it differs from 10-way digit classification. Figure S24 gives a visual summary of each benchmark.

Context gating. Inputs are contextual figure-ground MNIST examples. The right half of the image carries the original MNIST digit, while the left half is replaced by low-amplitude uniform noise; all pixels therefore remain nonnegative. This task tests whether credit assignment can exploit a simple context-dependent spatial structure without leaving the nonnegative-drive regime of the conductance model. The HSIC auxiliary objective (weight 0.01) is used to decorrelate layer representations, adding roughly 3 pp under local learning.

Noise resilience. Inputs are flattened MNIST images corrupted by structured channel noise: a fixed 50-dimensional latent Gaussian perturbation is projected into input space and added to each example, with $\sigma_{\text{task}} = 1.5$, then clipped to $[0,1]$. The projection is fixed across training, so the network must filter correlated interference to recover the underlying class signal. Credit assignment must remain coherent under the corrupted signal, making this task particularly sensitive to path-gain dispersion.

Info shunting. Inputs contain two additive streams: a low-frequency signal and a high-frequency distractor drawn with equal energy, together with a separate context channel indicating which stream is relevant on each trial. This benchmark uses signed synthetic inputs, so it violates the nonnegative presynaptic-drive assumption enforced in the main experiments. We therefore retain it only as an exploratory stress test and exclude it from the main-text claims.

Cue integration. Two noisy cue streams (A and B) are presented simultaneously, each carrying partial information about one of 2 classes. A binary one-hot context signal indicates which cue is reliable on each trial (cue A or cue B), and each cue vector is clipped to $[0,1]$ after noise is added. The fixed-pathway variant duplicates the context into both cue branches; the learned-routing variant requires the router to discover the cue separation from data. This task is used in Sec. A to test structured pathway-vector broadcast while staying inside the nonnegative-input regime.

J Depth Scaling and Noise Robustness

Depth scaling. Varying dendritic depth from 1–4 layers (branch factors [9] to $[3,3,3,3]$): shunting local degrades from 63.5% to 57.4%; additive local degrades more steeply from 54.9% to 29.7%. The shunting advantage grows with depth ($+8.5 \rightarrow +27.7$ pp). Backprop ceilings remain stable at about 90–92%. See Fig. S6A.

Noise robustness. Gaussian noise $\mathcal{N}(0, \sigma^2)$ on broadcast error: shunting is robust to $\sigma \leq 0.1$ (about 62%) while additive drops from 46.5% to chance at $\sigma = 1.0$, confirming that shunting credit signals carry useful learning information beyond the broadcast magnitude. See Fig. S6B.

Setting	Encoder	E layers	Tree	Synapses / branch	Seeds	Notes
Gradient fidelity / path-gain / noise-resilience mechanism	id.+ReLU	$[128,128]$	$[3,3]$	N_E=40, $N_I \in \{0,5,10,20,40\}$	5 fidelity, 5 oracle	Main mechanism sweeps in Figs. 2 and 3; reactivation `param_tanh`.
MNIST / context-gating local competence	id.+ReLU	$[128]$	$[3,3]$	N_E=40, N_I=20	5	Best local 5F per-soma runs in Table S4; reactivation `param_tanh`.
Fashion-MNIST competence	id.+ReLU	$[128]$	$[3,3]$	N_E=40, N_I=20	5	Dedicated five-seed standard-vs-local competence sweep; reactivation `param_tanh`.
Cue-routing PV-LocalCA	router	$[64]$	$[2]$	N_E=10, N_I=4	5	Learned router with two pathway groups and rank-2 structured feedback; reactivation `param_tanh`.
Positive cue-routing inhibition controls	router	$[64]$	$[2]$	N_E=10, N_I=4 direct I-to-E or no I-to-E	5 per condition	Strictly nonnegative router outputs; learned-I vs. no-I controls for rank-1, rank-2, pathway-vector, path transport, and BP.
Double-CIFAR routed screen	router	$[256]$	$[2]$	N_E=20, N_I=6 direct I-to-E or no I-to-E	5 per condition	Strictly nonnegative routed CIFAR streams; retained as a boundary screen because matched BP remains near chance.
CIFAR-10 compact direct-I-stream control ladder	id.+ReLU	$[20]$ E, no I cells	$[3,3,3,3]$	N_E=25, N_I=25 direct I-to-E, with matched no-I controls	5 per condition	Harder-data extension with input-driven inhibitory conductance, decoder $[32,16]$, decoder-aware soma mapping, and additive fairness controls.
Morphology×inhibition appendix map	id.+ReLU	$[128,128]$	$[2,2]$, $[3,3]$, $[4,4]$, $[2,2,2]$, $[3,3,3]$	N_E=40, $N_I \in \{0,5,10,20,40\}$	3	Supportive 3-seed sweep; reactivation `param_tanh`.

Table S16: **Representative architectures used in the manuscript.** All dendritic rows use explicit branch-level excitatory and inhibitory synapse banks with `somatic_synapses=false`, nonnegative conductances, and a nonnegative first-layer drive enforced by an explicit ReLU transfer stage in the main runs. The tree column gives dendritic morphology per excitatory neuron, and the synapse counts report excitatory and inhibitory synapses per branch.

Setting	Opt.	LR	Batch	Epochs	W-dec.	Notes
MNIST / Fashion-MNIST / context gating (5F per-soma LocalCA)	Adam	10^{-3} / $5 \cdot 10^{-4}$ (block / react.)	256	100	0	fixed-epoch, no early stopping; `param_tanh` reactivation
MNIST / Fashion-MNIST / context gating (BP ceiling)	Adam	10^{-3} / $5 \cdot 10^{-4}$	256	100	0	matched to LocalCA setup
Gradient fidelity / path-gain / noise resilience	Adam	10^{-3}	256	50	0	5 seeds, hooks enabled for diagnostic capture
Morphology $\times$ inhibition regime map (exploratory)	Adam	10^{-3}	256	50	0	3 seeds
Cue routing (soma-off & soma-on)	Adam	10^{-3}	256	90	0	early stopping with patience 30
Positive cue-routing inhibition controls	Adam	$1.5 \cdot 10^{-3}$ / $7 \cdot 10^{-4}$ (block)	256	140	0	early stopping with patience 35; 50 complete runs
Double-CIFAR routed screen	Adam	$1.2 \cdot 10^{-3}$ / $6 \cdot 10^{-4}$ (block)	256	220	0	no early stopping; 70 complete runs; retained as boundary screen
CIFAR-10 compact direct-I-stream depth-4 LocalCA	Adam	10^{-3} / 10^{-4} (block / react.)	256	400	0 / 0.01	grad-clip 5.0, early stop patience 50, nonlinear decoder
CIFAR-10 compact direct-I-stream depth-4 BP	Adam	10^{-3}	256	200	0.01	early stop patience 40, matched ceiling
Weight-distribution vs. depth	Adam	10^{-3}	256	100	0	`weight_analysis` hook captures final-epoch statistics
Component ablation (MNIST)	Adam	10^{-3}	256	100	0	5 seeds, single-layer dendrinet $[3,3]$
(b,m) policy comparison (MNIST / CIFAR-10)	Adam	10^{-3} / 10^{-4}	256	100 / 400	0 / 0.01	quantile refresh every 5 epochs; EMA $\alpha \in \{0.25, 0.5, 1.0\}$

Table S17: **Training hyperparameters by experiment.** All runs use Adam with default β_1=0.9, β_2=0.999, ε=10^{-8}. Reactivation and block-linear parameter groups use a slower LR to keep conductance-level dynamics stable. Full configs are provided under `drafts/dendritic-local-learning/configs/`.

Setting	Hidden width(s)	Output dim	Effective default per-soma field
Gradient fidelity / path-gain / noise resilience	$128, 128$	10	Scalar / rank-1 at both dendritic layers
MNIST / Fashion-MNIST competence	128	10	Scalar / rank-1 at the dendritic hidden layer
Context gating competence	128	2	Scalar / rank-1 at the dendritic hidden layer
Cue routing	64	2	Scalar / rank-1 by default; higher-rank tests use explicit low-rank or pathway-vector feedback
Compact CIFAR-10 harder-data extension	20	decoder-input dim 20	Vector-valued per-soma after decoder-aware mapping; higher-rank tests use explicit low-rank or path transport

Table S18: **Effective dimensionality of the default "per-soma" broadcast in the headline architectures.** In code, per-soma uses the output error vector only when layer width matches decoder output dimension; otherwise it falls back to a shared scalar. The main feedforward settings in this paper therefore test a genuinely rank-1 broadcast regime by default, which is why the higher-rank low-rank, path-transport, and pathway-vector controls are important.

Extension	Meaning	Compatible with	Typical use here
Per-soma broadcast	Somatic error vector when widths match, otherwise scalar fallback	3F, 4F, 5F	Default main-text LocalCA setting; in the headline feedforward runs this is effectively rank-1
Low-rank broadcast	Small number of broadcast channels instead of a scalar or full field	3F, 4F, 5F	Intermediate control when rank-1 is too restrictive
Path transport	Recursive tree-transported error using conductance-stage path gains	3F, 4F, 5F	Oracle upper-bound broadcast
Path propagation	Approximate path-gain attenuation applied to broadcast	3F, 4F, 5F	Morphology-aware modifier for deeper trees
Pathway vector	Structured branch/pathway-specific broadcast field	3F, 4F, 5F	Used mainly in PV-LocalCA on routed tasks
Branch-role rules	Scales updates by router- or pathway-derived branch specialization	3F, 4F, 5F	Used in the structured pathway extension
Inhibitory homeostasis	Auxiliary local inhibitory balancing term	3F, 4F, 5F	Optional stabilizer in routed or shunting settings

Table S19: Compatibility of the newer LocalCA extensions with the base 3F/4F/5F family. These extensions mostly modify the error field or branch scaling rather than defining a new “6F” or “7F” rule.

Figure / Table	Sweep	Replication	Tier
Fig. 1	Model, rule, and broadcast definitions	N/A	Definitions
Fig. 2	Gradient-fidelity diagnostic	5 seeds	Headline
Fig. S3	Alignment dynamics and nonzero-gradient check	3 seeds	Supportive
Fig. 3	Mechanistic chain + oracle + summary	5 seeds	Headline
Fig. 4	Competence / regime dose-response	5 seeds	Headline
Fig. S9 / Table S6	CIFAR-10 compact direct-I-stream control ladder	5 seeds per condition; 70 complete runs	Supportive
Fig. S16	Cue-routing (soma off)	5 seeds	Supportive
Table S4	Competence gaps (BP ceiling vs. 5F)	5 seeds	Headline
Fig. S17	Soma-on extension across manuscript families	Family aggregates (144–420 cfgs)	Supportive
Table S13	Soma-on delta summary across manuscript families	Family aggregates	Supportive
Fig. S19	(b,m) update-policy comparison	Matched 4-cell grid / bar	Supportive
Fig. S20	Component ablation (MNIST)	5 seeds	Supportive
Fig. S18	Cue-routing soma-on (stable subset)	5 seeds	Supportive
Table S9	Fitted direct-I inhibitory path-gain diagnostic	5 runs	Supportive
Fig. S14	Positive cue controls and routed Double-CIFAR screen	50 cue runs; 70 Double-CIFAR runs	Supportive / boundary
Fig. S15	Inhibition interventions and exact-error rank	5 seeds	Supportive
Fig. S21	Weight-distribution sensitivity	5 seeds	Supportive
Fig. S22	Weight-distribution vs. depth	5 seeds	Supportive
Fig. S23	Weight-distribution soma-off vs. soma-on	5 seeds	Supportive
Fig. S10 / Table S7	CIFAR-10 depth-4 soma extension	3 seeds	Supportive
Fig. S8	FA/DFA baseline comparison	5 seeds	Supportive
Fig. S7	Broadcast-bandwidth / quantization	5 seeds	Supportive
Fig. S11	Additive + normalization control	5 seeds	Supportive
Fig. S6 / S5	Stress tests; verification	5 seeds	Supportive
Fig. S4	5F sensitivity	3 seeds	Supportive
Table S10	Rank-bridge	5 seeds	Supportive
Fig. S12	Morphology $\times$ inhibition map	3 seeds	Supportive
Fig. S13 / Table S8	One-layer direct-I vs. explicit-I input-mode probe	3 seeds	Supportive
Signed-input info-shunting	Signed-input stress test	1–3 seeds	Exploratory

Table S20: **Evidence map.** Each main-paper and appendix figure or table is pinned to its sweep and replication type. Most main claims come from dedicated 5-seed sweeps; the supporting soma-extension and (b,m) policy appendices instead summarize matched sweep families or matched config grids, and are labeled accordingly. The BP ceilings in Table S4 come from dedicated matched-capacity 5-seed standard-training refreshes.

Asset	Use in this paper	Access / license note
MNIST	Digit classification and derived nonnegative synthetic tasks	Public academic benchmark credited to the original dataset authors; used through standard public access, with no raw-data redistribution.
Fashion-MNIST	Apparel classification stress test	Public benchmark released with the Fashion-MNIST paper and repository; used through standard public access, with no raw-data redistribution.
CIFAR-10	Flattened harder-data diagnostic	Public research benchmark from the CIFAR dataset collection; used through standard public access, with no raw-data redistribution.
MICrONS	Biological reference statistics in the weight-distribution appendix	Public connectomics resource cited to the MICrONS Consortium; only aggregate comparison statistics are reported here.
Synthetic tasks	Context gating, noise resilience, cue integration	Generated procedurally from public benchmarks or random seeds described in Appendix I; no new third-party asset is introduced.

Table S21: **Existing assets used in the manuscript.** We credit the original sources in the bibliography, use public benchmark assets under their published access conditions, and do not redistribute third-party raw data in the anonymous submission.

Rule	Factors	Cost	Best regime
3F	$x, (E-V), e$	$\mathcal{O}(1)$	Baseline
4F	3F $+\rho$	$\mathcal{O}(1)$	Improved conditioning
5F	4F $+\phi$	$\mathcal{O}(d_n)$	Strongest overall
5F+VH	5F + local voltage homeostasis	$\mathcal{O}(d_n)$	Stabilized LocalCA / warm-up
PV-LocalCA	3F + structured e_n + role alignment	$\mathcal{O}(r_n)$	Pathway discovery

Table S22: Variant taxonomy. ‘5F+VH‘ denotes the same base 5F rule with an additional strictly local voltage-centering auxiliary that can also be scheduled through multi-stage LocalCA warm-up; the codebase also supports this as a shorthand ‘5f_vh‘ alias in configuration files.

Component	Analog	Interpretation
R_n^{tot}	Input resistance	Sensitivity modulation
$(E_j - V_n)$	Synaptic driving force	Local gradient factor
Shunting	Divisive normalization	$\partial V / \partial g_I \propto -V$
ρ_n	Layer relevance	Output correlation
ϕ_n	Signal propagation	Conditional predictability
Pathway vector	Branch-specific top-down drive	Structured teaching signal
Role alignment	Pathway-selective plasticity	Branch specialization

Table S23: Biological analogs.

	MNIST		Fashion-MNIST	
Condition	Exc σ	Inh σ	Exc σ	Inh σ
BP + Shunting	0.94±0.01	0.87±0.01	0.92±0.01	0.84±0.01
BP + Additive	1.30±0.01	1.22±0.01	1.16±0.01	1.05±0.01
Local + Shunting	1.22±0.01	1.71±0.02	1.13±0.06	1.36±0.06
Local + Additive	1.69±0.16	1.57±0.07	1.43±0.16	1.11±0.12
Song et al. 2005	$\sigma = 0.94$ (rat V1 L5, EPSP amplitude)			
MICrONS E→E	$\sigma = 1.14$ (mouse V1, synapse size, n=3.9M)			
MICrONS I→E	$\sigma = 0.92$ (mouse V1, synapse size, n=3.5M)			

Table S24: **Weight distribution width.** Log-normal σ of excitatory and inhibitory synaptic weights (mean±std across 5 seeds). Biological references are included only as descriptive anchors. With biologically constrained E/I ratios (4:1), shunting self-normalization produces weight heterogeneity in the range spanned by those references, whereas additive integration is broader and less stable.